# SlimDiffSR: Toward Efficient Remote Sensing Image Super-Resolution via Diffusion Model Distillation

Ce Wang, Zhenyu Hu, and Wanjie Sun

***Abstract*—Diffusion models (DMs) have recently achieved remarkable performance in image super-resolution (SR), but their high computational cost limits practical deployment in remote sensing applications. To address this issue, we propose SlimDiffSR, an efficient diffusion-based framework for real-world remote sensing image SR. Unlike existing single-step diffusion methods that rely on fixed timesteps, we first introduce an uncertainty-guided timestep assignment strategy to construct a stronger single-step teacher model, where reconstruction difficulty is explicitly linked to diffusion timesteps, enabling adaptive generative strength. Building upon this teacher, we further present a structured pruning strategy tailored to remote sensing imagery, which systematically removes redundant semantic modules and replaces standard operations with lightweight designs, including frequency-separable convolution (FSC), direction-separable convolution (DSC), and a query-driven global aggregation module (QGAM). These components explicitly exploit the unique characteristics of remote sensing data, such as sparse high-frequency details, strong directional patterns, and long-range spatial dependencies. To strengthen the distillation process, we introduce maximum mean discrepancy (MMD)-based distribution alignment alongside mean squared error (mse) feature supervision, improving peak signal-to-noise ratio (PSNR) by 0.23 dB. Extensive experiments on multiple remote sensing benchmarks demonstrate that SlimDiffSR achieves a favorable balance between efficiency and reconstruction quality. In particular, it attains up to 200× inference acceleration and a 20× reduction in model parameters compared with multistep DMs, while achieving competitive perceptual quality and clearly outperforming existing efficient diffusion baselines in efficiency. The code is available at https://github.com/wwangcece/SlimDiffSR**

***Index Terms*—Diffusion model (DM), efficient model, image super-resolution (SR), remote sensing.**

## I. Introduction

HIGH-RESOLUTION (HR) remote sensing imagery is critical for a wide range of Earth observation and monitoring applications, including but not limited to weather forecasting [1], object detection [2], and land use and land cover mapping [3]. However, the spatial resolution of many satellite images is inherently constrained by hardware limitations and transmission bandwidth, often resulting in insufficient detail for reliable interpretation and analysis [4]. To address this limitation, super-resolution (SR) techniques aim to enhance the spatial fidelity of low-resolution (LR) observations through data-driven reconstruction, enabling the generation of HR imagery in a cost-effective manner.

Over the past decades, a variety of SR approaches have been developed [5], including reconstruction-based methods, sparse representation-based methods, and regression-based methods. More recently, deep learning has significantly advanced SR performance by leveraging large-scale datasets to learn complex LR-to-HR mapping. Early deep SR research primarily focused on network architecture design [6], [7], [8], loss functions [9], [10], and optimization strategies [5], achieving substantial improvements in distortion-oriented metrics such as peak signal-to-noise ratio (PSNR). Subsequently, generative adversarial networks (GANs) [11], [12] were introduced to enhance perceptual quality by encouraging the generation of realistic textures through adversarial training. Despite their success, GAN-based methods often suffer from training instability, mode collapse, and the introduction of undesirable visual artifacts [13]. In recent years, diffusion models (DMs) [14] have demonstrated remarkable success in high-fidelity image generation and have been extended to a variety of low-level vision tasks, including image-to-image translation and image restoration. Their application to SR has shown promising results, offering improved perceptual quality, enhanced training stability, and more diverse reconstruction outcomes [15], [16], [17].

Despite these advantages, diffusion-based SR methods remain computationally expensive due to their iterative sampling procedures and large model sizes. For instance, a typical multistep DM [16] may require more than 100 s to process a single 1024 × 1024 image on an RTX 3090 GPU. Such computational cost poses a significant barrier to large-scale deployment in remote sensing applications, where massive volumes of imagery must be processed efficiently. To mitigate this issue, recent studies have focused on developing efficient diffusion frameworks, which can be broadly categorized into three directions. First, task-specific diffusion formulations have been proposed for SR and image restoration [15], [18], aiming to exploit degraded inputs more effectively while

Received 18 May 2026; revised 22 July 2026 and 26 August 2026; accepted 8 September 2026. Date of publication 11 September 2026; date of current version 22 September 2026. This work was supported in part by the National Natural Science Foundation of China under Contract 62571377 and Contract 62201403 and in part by Hubei Provincial Natural Science Foundation of China under Contract 2024AFB549. *(Corresponding author: Wanjie Sun.)*

The authors are with the School of Remote Sensing and Information Engineering, Wuhan University, Wuhan, Hubei 430072, China (e-mail: sunwanjie@whu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2026.3732853

1558-0644 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 14:26:39 UTC from IEEE Xplore. Restrictions apply.

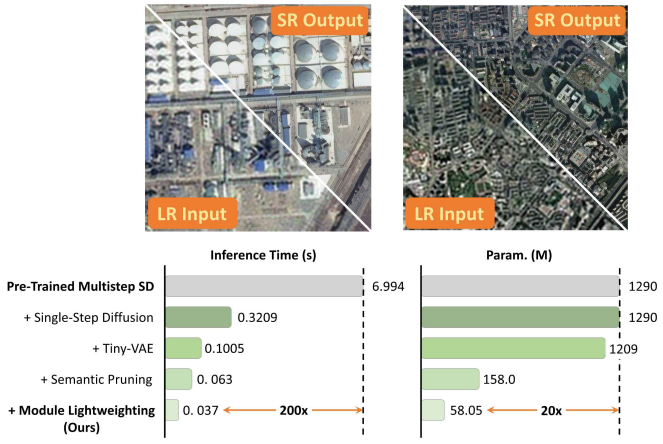



Fig. 1. Our method achieves a substantial reduction in both inference time and model size. The inference speed is evaluated by processing a single 512 × 512 image on an RTX 3090 GPU. By constructing a single-step DM and further incorporating structured pruning techniques, the final efficient model achieves up to 200× faster inference and a 20× reduction in the number of parameters compared with the original SD 2.1 model, while maintaining competitive restoration performance.

reducing the number of denoising steps. Second, advanced ordinary differential equation (ODE) and stochastic differential equation (SDE) solvers have been introduced to accelerate the sampling process [19], [20]. Third, model compression techniques, such as knowledge distillation and structured pruning, have been employed to simultaneously reduce sampling steps and model complexity [21], [22].

Despite these advances, several challenges remain when applying efficient DMs to remote sensing SR, as follows.

1) *Fixed Timestep:* Existing single-step DMs typically adopt a fixed timestep [21], [23], ignoring the fact that different timesteps in multistep diffusion pretraining provide distinct guidance signals for the denoising process.
2) *Suboptimal Pruning Strategies:* Current pruning methods are largely adapted from natural image generation tasks [22], without accounting for the unique characteristics of SR and remote sensing imagery.
3) *Limited Distillation Objectives:* Most existing distillation approaches rely on mean squared error (mse) to align intermediate features between teacher and student networks [22], which only captures second-order statistics and fails to model distribution-level discrepancies.

To address these challenges, we propose SlimDiffSR, an efficient diffusion-based framework for real-world remote sensing image SR. As illustrated in Fig. 1, the proposed method achieves significant improvements in both inference efficiency and model compactness. The overall framework consists of two stages: teacher model training and student model distillation.

In the first stage, we construct an unpruned single-step diffusion-based SR network as the teacher model. Although existing single-step diffusion-based methods, such as OSEDiff [23] and S3Diff [24], demonstrate promising runtime efficiency, they rely on a fixed diffusion timestep and thus fail to fully exploit timestep-dependent guidance learned during diffusion pretraining. Our prior studies [25] have shown that different inputs may require different timesteps to achieve better restoration quality due to varying degradation levels. Motivated by this observation, we propose an uncertainty-guided single-step DM for remote sensing image SR as the teacher model. Specifically, we establish a correspondence between reconstruction uncertainty and diffusion timesteps. An uncertainty estimator is first trained in the latent space using an uncertainty-aware objective [26] to predict reconstruction difficulty. A timestep inversion strategy is then introduced to map uncertainty scores to appropriate diffusion timesteps, enabling dynamic timestep assignment for different LR inputs.

In the second stage, the teacher model is pruned and distilled to improve inference efficiency while preserving reconstruction performance. This stage consists of two steps: structural pruning and distillation training. For the pruning strategy, the denoising network is decomposed into removable and replaceable modules. Since LR inputs already contain substantial semantic information, and deeper UNet layers of the DM primarily capture high-level semantics [27], these components are considered redundant for SR and can be removed. Second, considering the characteristics of remote sensing imagery, we redesign the remaining modules to better match the data properties. Remote sensing images typically contain large homogeneous regions with limited low-frequency variation, while critical information is concentrated in high-frequency structures such as edges and object boundaries. To exploit this, we introduce frequency-separable convolution (FSC) to emphasize informative high-frequency components while reducing redundant computation. In addition, remote sensing scenes often exhibit strong directional patterns (e.g., roads and rivers). To capture such anisotropic structures efficiently, we design direction-separable convolution (DSC). Furthermore, due to the large spatial extent of remote sensing images, long-range dependencies are important but costly to model with standard self-attention. Therefore, we propose a query-driven global aggregation module (QGAM) to approximate global context in a more efficient manner. These designs replace standard convolution and self-attention layers, improving efficiency while preserving task-relevant information. For distillation, a multiscale feature supervision strategy is adopted to transfer knowledge from the teacher to the student network. To address the distribution mismatch between their features, we incorporate the maximum mean discrepancy (MMD) loss [28] with mse loss to enforce both pointwise and distribution-level alignment.

Unlike existing efficient diffusion-based SR methods that directly approximate multistep inference with fixed designs, this work addresses a key limitation in single-step diffusion for remote sensing SR, i.e., the lack of adaptive generative strength under varying degradation levels. To this end, we introduce an uncertainty-guided timestep assignment strategy to construct a stronger single-step teacher model, where reconstruction difficulty is explicitly linked to diffusion timesteps. Building upon this teacher, we further develop an efficient student model through RS-aware structural redesign. Instead of applying generic pruning strategies, we exploit the characteristics of remote sensing imagery to

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 14:26:39 UTC from IEEE Xplore. Restrictions apply.

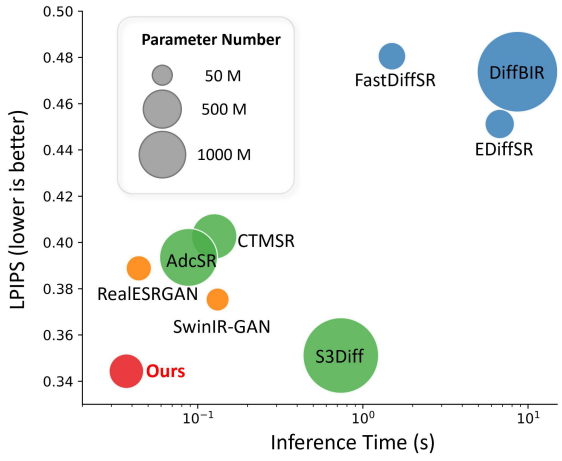



Fig. 2. Performance and efficiency comparison among different remote sensing image SR methods for processing a single 512×512 image on an RTX 3090 GPU. Blue bubbles denote multistep diffusion-model-based methods, green bubbles denote single-step diffusion-model-based methods, and yellow bubbles denote GAN-based methods.

remove redundant semantic modules and replace remaining components with efficient alternatives. This teacher–student design enables us to improve reconstruction quality through adaptive diffusion guidance, while achieving high inference efficiency via efficient model construction. As shown in Fig. 2, while maintaining a compact model size and strong reconstruction quality, our efficient model not only significantly outperforms existing multistep and single-step DM-based methods in inference speed, but also surpasses GAN-based approaches.

The main contributions of this work are as follows.

1) We identify the limitation of fixed-timestep selection in existing single-step diffusion SR models and propose an uncertainty-guided timestep assignment strategy. By explicitly linking reconstruction difficulty with diffusion timesteps, the proposed method enables adaptive generative strength and improves the effectiveness of single-step diffusion for remote sensing SR.
2) We show that effective model compression for remote sensing SR requires RS-specific structural redesign rather than generic pruning. Based on this insight, we remove redundant deep semantic modules and introduce efficient operators, including FSC, DSC, and query-driven global aggregation, which better match the characteristics of remote sensing imagery.
3) We develop a two-stage teacher–student diffusion framework that decouples quality enhancement from efficient deployment. The adaptive teacher generates high-quality LR-specific SR features under different generative levels, while the student learns a direct LR-to-SR mapping through feature distillation and performs single-step inference without additional timestep prediction.
4) We incorporate MMD into the distillation process to complement conventional mse supervision, enabling alignment of both point-wise features and their underlying distributions, which leads to improved perceptual quality and reconstruction fidelity.

## II. Related Work

### A. Remote Sensing Image SR

Early research on remote sensing image SR predominantly relied on convolutional neural networks (CNNs). Representative models such as SRCNN [29] and VDSR [7] were directly applied to satellite imagery, demonstrating the effectiveness of deep learning for enhancing both panchromatic and multispectral data. For example, Tuna et al. [30] showed that VDSR performs well on HR images from SPOT and Pleiades sensors. However, these early models are relatively shallow and lack sufficient contextual modeling capability, limiting their ability to capture complex spatial patterns in remote sensing scenes.

To address this limitation, deeper and task-specific CNN-based models have been proposed. The RS-DRL network [31] introduces residual learning for Sentinel-2 data, while the deep memory connected network (DMCN) [32] incorporates local and global memory mechanisms to enhance multilevel feature representation. Although these methods achieve strong performance in distortion-oriented metrics such as PSNR and structural similarity index (SSIM), they often produce over-smoothed textures and fail to preserve fine details.

To improve perceptual quality, GANs have been introduced into remote sensing SR. Models such as ESRGAN [33], EEGAN [34], DRGAN [35], and SRADSGAN [36] employ adversarial training and perceptual losses to generate sharper and more realistic images. In addition, task-specific variants such as SRAGAN [37] are designed to better capture structural characteristics in remote sensing imagery. However, GAN-based methods often suffer from inferior quantitative performance and may introduce artifacts, especially under large scaling factors or cross-sensor scenarios. To mitigate these issues, subsequent studies have explored hybrid loss functions [38] and improved discriminator designs [39] to balance perceptual quality and reconstruction fidelity.

### B. SR DMs

DMs have recently emerged as a powerful paradigm for image SR, initially in the domain of natural images. Early approaches such as SR3 [40] and SRDiff [41] train DMs from scratch to reconstruct HR images from LR inputs, but require substantial computational resources and large-scale paired datasets. To alleviate this limitation, recent methods leverage pretrained text-to-image DMs as strong generative priors. For example, DiffBIR [17] and StableSR [16] utilize pretrained Stable Diffusion (SD) [42] to guide texture synthesis, demonstrating that pretrained generative knowledge can be effectively transferred to SR tasks.

Despite their strong performance, pretrained DMs are characterized by large parameter sizes and slow inference speeds, which hinder practical deployment. To improve efficiency, recent studies have explored efficient diffusion techniques from three main perspectives. First, task-specific diffusion formulations, such as ResShift [15] and RDDM [18], aim to reduce the number of denoising steps. Second, efficient ODE/SDE-based samplers [19], [20] are developed to accelerate the sampling process. Third, model compression

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 14:26:39 UTC from IEEE Xplore. Restrictions apply.

techniques, including knowledge distillation and structured pruning, are employed to jointly reduce sampling steps and model complexity. For instance, OSEDiff [23] and S3Diff [24] achieve single-step inference via distillation, while AdcSR [21] reduces the model size through structured pruning. LinearSR [43] further introduces linear attention into diffusion-based SR frameworks to accelerate inference.

In the domain of remote sensing, SR poses unique challenges due to complex textures, large-scale variations, and sensor-specific imaging characteristics. Early DM-based SR methods for remote sensing typically relied on training models from scratch using domain-specific datasets [44]. For instance, EDiffSR [45] introduces an efficient diffusion-based framework for remote sensing SR under fixed degradation assumptions. DiffBSR [46] further extends DMs to blind SR by introducing a conditional diffusion framework with explicit degradation representation to handle unknown and complex degradations. CNN-DiffSR [47] explores a CNN-guided diffusion paradigm, where an initial SR result produced by a CNN is used as conditional guidance for iterative diffusion-based refinement, improving reconstruction quality and visual fidelity. More recent studies have attempted to bridge the gap between natural-image diffusion foundation models and remote sensing applications. SGDM [48], for instance, leverages pretrained SD models to perform 16× and 32× SR for remote sensing imagery by disentangling style and content representations and incorporating vector map priors to enhance semantic consistency. Likewise, DiffusionSat [49], the first diffusion foundation model for remote sensing, combines pretrained SD with auxiliary metadata (e.g., geographic coordinates) to support temporal SR and image inpainting, demonstrating superior performance over GAN-based approaches.

Despite these advances, existing diffusion-based SR methods for remote sensing remain computationally expensive and are not well suited for large-scale deployment. Meanwhile, efficient diffusion techniques developed for natural images do not fully consider the unique characteristics of remote sensing data, as discussed in Section I. Therefore, there remains a need for a dedicated efficient diffusion framework tailored to remote sensing image SR.

### C. Remote Sensing Image Interpretation

Recent advances in remote sensing interpretation have demonstrated the increasing demand for high-quality remote sensing imagery to support fine-grained analysis tasks. In particular, deep learning-based methods have achieved remarkable progress by exploiting hierarchical features and contextual information. For instance, CT-HiffNet [50] introduced a contour-texture hierarchical feature fusion strategy to enhance boundary and texture representation for cropland field parcel extraction from HR imagery. Moreover, transformer-based architectures such as SegFormer [51] have been investigated for semantic segmentation of UAV remote sensing images, demonstrating the effectiveness of attention mechanisms in capturing complex spatial patterns, while FPENet [52] achieves an accuracy–efficiency tradeoff through lightweight pyramid context encoding and feature embedding. Similarly,

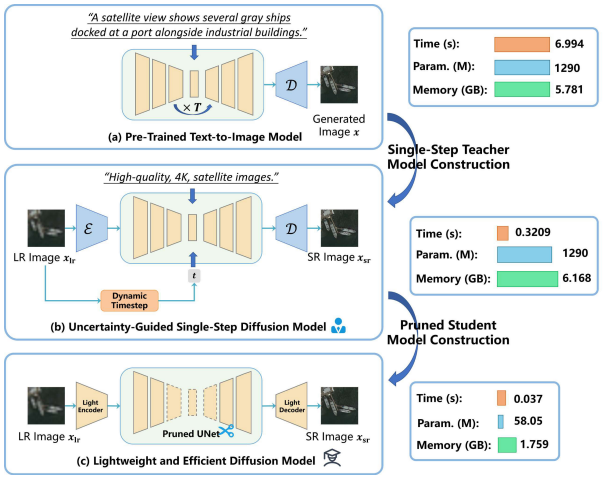



Fig. 3. Overview of the proposed SlimDiffSR framework. In the first stage, an uncertainty-guided single-step diffusion teacher is constructed to generate high-quality features for supervision. In the second stage, the intermediate features produced by the adaptive teacher are distilled into an efficient student model, where redundant modules are removed or replaced. The final student directly maps the LR input to the SR output through timestep-free single-step inference.

CTHBNet [53] developed a CNN-Transformer hybrid architecture with boundary guidance to effectively capture both local details and global contextual dependencies for cropland parcel mapping. For the related task of terraced-field recognition, JAM-R-CNN [54] combines a jumping network and task-adaptive anchors to improve the identification of narrow and elongated targets. For medium-resolution satellite imagery, S2-IFNet [55] proposed a spatial-semantic information fusion framework with boundary feature enhancement for forest land extraction from Sentinel-2 data. Beyond semantic interpretation, GPDRNet [56] preserves the geometry of typical ground objects during multimodal image registration under local nonrigid deformation. Beyond task-specific architectures, recent studies have also explored the potential of foundation models for remote sensing interpretation. FlexiMo [57] presented a flexible remote sensing foundation model capable of adapting to diverse downstream tasks through large-scale pretraining. These developments highlight that improving the spatial fidelity and semantic consistency of remote sensing images is crucial for subsequent interpretation tasks, which further motivates the development of efficient and high-quality remote sensing image SR methods.

## III. Method

### A. Overview

The overall framework of SlimDiffSR is illustrated in Fig. 3. The goal of this work is to develop an efficient diffusion-based model for remote sensing image SR that achieves high reconstruction quality while maintaining efficient inference. To this end, we adopt a two-stage teacher–student framework. In the first stage, we construct a strong single-step diffusion teacher model to provide high-quality supervision. Unlike existing single-step diffusion methods that rely on fixed timesteps, we introduce an uncertainty-guided timestep assignment strategy to enhance the reconstruction capability of the teacher model

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 14:26:39 UTC from IEEE Xplore. Restrictions apply.

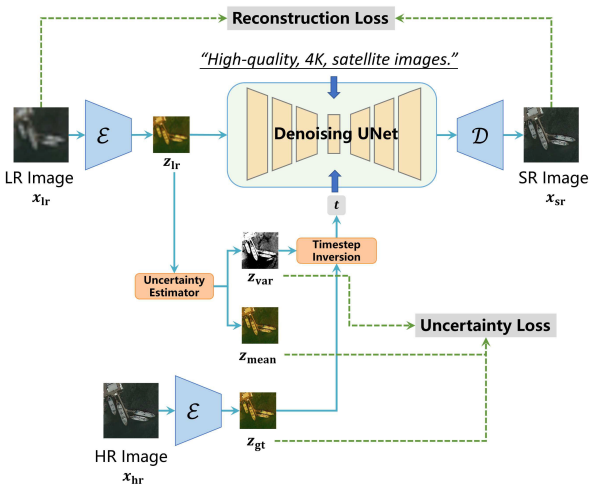



Fig. 4. Overview of the uncertainty-guided single-step DM in the first stage.

TABLE I
NOTATION SUMMARY FOR THE SINGLE-STEP TEACHER MODEL CONSTRUCTION

| Symbol | Definition |
|---|---|
| $x_{\mathrm{lr}}$ | Low-resolution input image. |
| $z_{\mathrm{lr}}$ | Latent representation of $x_{\mathrm{lr}}$. |
| $z_{\mathrm{gt}}$ | Ground-truth high-resolution latent representation. |
| $z_{\mathrm{mean}}$ | Predicted mean latent from uncertainty estimator. |
| $z_{\mathrm{var}}$ | Predicted element-wise uncertainty (variance) map. |
| $\mathrm{UE}_{\phi}$ | Uncertainty estimator with parameters $\phi$. |
| $\mathrm{Pred}_{\theta}$ | Noise-level predictor with parameters $\theta$. |
| $\bar{\alpha}_t$ | Diffusion cumulative noise schedule at timestep $t$. |
| $\bar{\alpha}_{\mathrm{target}}$ | Predicted target noise level for adaptive guidance. |
| $t,\ t^*$ | Original diffusion timestep and inferred adaptive timestep. |
| $\sigma_{\mathrm{gt}}^2$ | Signal variance of the HR latent distribution. |
| $\epsilon$ | Gaussian noise, $\epsilon \sim \mathcal{N}(0, \mathrm{I})$. |

under varying degradation levels. In the second stage, we focus on efficiency by designing an efficient student model through RS-aware structural pruning and module replacement tailored to remote sensing imagery.

Importantly, the adaptive timestep mechanism is only used during teacher model construction. The final student model operates with a fixed single-step inference, introducing no additional computational overhead. During distillation, the adaptive teacher generates LR-specific intermediate features and SR predictions using generative levels selected according to the corresponding LR-HR training pair. The student receives only the LR input and is optimized to match these teacher features while minimizing the reconstruction objective. Because the student features remain conditioned on the LR input, the student model can produce different SR behaviors for LR inputs exhibiting different contents and degradations without introducing an additional timestep prediction. Through this design, the proposed framework effectively decouples reconstruction quality and inference efficiency, enabling SlimDiffSR to achieve a favorable balance between performance and efficiency for large-scale remote sensing applications.

### B. Single-Step Teacher Model Construction

The objective of the teacher model is to provide strong reconstruction supervision for the subsequent efficient student. A key limitation of existing single-step diffusion methods lies in their use of a fixed timestep, which implicitly assumes a uniform degradation level across all inputs. However, in remote sensing SR, degradation varies across scenes and spatial positions, making such a fixed design suboptimal.

To address this issue, we introduce an uncertainty-guided timestep assignment strategy to construct a stronger single-step teacher model. Instead of treating the timestep as a fixed hyperparameter, we learn an adaptive diffusion guidance level conditioned on reconstruction uncertainty and latent structural information. The uncertainty map provides an important cue for estimating restoration difficulty, while the final timestep is optimized by the teacher restoration objective. As illustrated in Fig. 4, given an LR input $x_{\mathrm{lr}}$, an uncertainty estimator first predicts an element-wise uncertainty map $z_{\mathrm{var}}$, which reflects reconstruction ambiguity. Then, a timestep inversion strategy estimates the corresponding diffusion timestep $t^*$ by jointly considering uncertainty-aware cues and latent structural information. The predicted timestep is used to guide the UNet denoising process, allowing the teacher model to adapt its generative strength to different inputs and provide stronger supervision than a plain fixed-timestep single-step diffusion SR model.

For clarity, the notations used in the following derivations are summarized in Table I.

*1) Uncertainty Estimation:* According to existing studies on uncertainty learning [26], we model the conditional distribution of the ground-truth latent $z_{\mathrm{gt}}$ given the LR latent $z_{\mathrm{lr}}$ as a Gaussian distribution with element-wise variance

$$z_{\mathrm{gt}} = z_{\mathrm{mean}} + \epsilon \cdot \sqrt{z_{\mathrm{var}}}, \quad \epsilon \sim \mathcal{N}(0, \mathrm{I}) \tag{1}$$

where $z_{\mathrm{mean}}$ and $z_{\mathrm{var}}$ denote the predicted mean and variance, respectively. They are estimated by the uncertainty estimator $\mathrm{UE}_{\phi}$

$$(z_{\mathrm{mean}}, z_{\mathrm{var}}) = \mathrm{UE}_{\phi}(z_{\mathrm{lr}}). \tag{2}$$

The uncertainty estimator plays two complementary roles. First, it produces a preliminary restoration $z_{\mathrm{mean}}$, which represents the expected HR latent under the predicted Gaussian distribution. Second, it predicts an element-wise variance map $z_{\mathrm{var}}$, which reflects reconstruction uncertainty. In general, texture-rich or structurally complex regions are more difficult to reconstruct and therefore tend to exhibit larger uncertainty values [58].

To train the estimator, we maximize the posterior probability of the predicted variance conditioned on the observed LR-HR latent pair. Since $z_{\mathrm{mean}}$ and $z_{\mathrm{var}}$ are both determined by $z_{\mathrm{lr}}$ through $\mathrm{UE}_{\phi}$, Bayes' rule gives

$$p(z_{\mathrm{var}} \mid z_{\mathrm{gt}}, z_{\mathrm{lr}}) \propto p(z_{\mathrm{gt}} \mid z_{\mathrm{mean}}, z_{\mathrm{var}})\, p(z_{\mathrm{var}}) \tag{3}$$

under the Gaussian observation model in (1), the likelihood is formulated as

$$p(z_{\mathrm{gt}} \mid z_{\mathrm{mean}}, z_{\mathrm{var}}) = \frac{1}{\sqrt{2\pi z_{\mathrm{var}}}} \exp\left(-\frac{\|z_{\mathrm{gt}} - z_{\mathrm{mean}}\|_2^2}{2 z_{\mathrm{var}}}\right). \tag{4}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 14:26:39 UTC from IEEE Xplore. Restrictions apply.

According to Jeffreys' prior [59], the uncertainty prior can be expressed as $p(z_{\text{var}}) \propto 1/z_{\text{var}}$. By taking the logarithm of the posterior and removing constant terms, we obtain

$$\ln p\left(z_{\text{var}} \mid z_{\text{gt}}, z_{\text{lr}}\right) \propto -\frac{\left\| z_{\text{gt}} - \text{UE}_{\phi}\left(z_{\text{lr}}\right)_{\text{mean}} \right\|_2^2}{2z_{\text{var}}} - \frac{3}{2}\ln z_{\text{var}}. \quad (5)$$

Minimizing the negative log-posterior leads to the following uncertainty-aware loss. Let $s = \ln z_{\text{var}}$ for numerical stability, then the loss is defined as

$$\mathcal{L}_{\text{uncertainty}} = \frac{1}{2N}\sum_{i=1}^{N}\left(\exp\left(-s^i\right)\left\| z_{\text{gt}}^i - z_{\text{mean}}^i \right\|_2^2 + 3s^i\right) \quad (6)$$

where $N$ denotes the number of latent elements. Here, the variance is predicted element-wise over the latent representation, and each latent element is modeled as an independent scalar Gaussian variable. This formulation enables the estimator to adaptively assign higher uncertainty to regions that are more difficult to reconstruct, which subsequently guides the timestep assignment in the DM.

*2) Timestep Inversion:* Our previous studies have shown that, for single-step DMs, inputs with different reconstruction difficulties should be guided by different denoising timesteps [25]. Given the uncertainty map estimated in Section III-B1, our goal is to determine an appropriate timestep for the teacher model, so that the generative strength of the DM can be adaptively adjusted according to the reconstruction ambiguity.

A straightforward solution is to directly predict the timestep $t$ with a neural network. However, this design is not suitable for end-to-end optimization, because the predicted timestep is used as a discrete index to retrieve the corresponding noise schedule value $\bar{\alpha}_t$. Such an indexing operation is nondifferentiable, making it difficult to train the timestep prediction module jointly with the teacher model. Therefore, instead of directly predicting the discrete timestep $t$, we predict a continuous noise-level variable $\bar{\alpha}_{\text{target}}$, which can be regarded as a continuous representation of the diffusion timestep.

To provide a principled target for predicting $\bar{\alpha}_{\text{target}}$, we establish the relationship between reconstruction uncertainty and diffusion noise level through the signal-to-noise ratio (SNR). In the forward diffusion process, the noisy latent at timestep $t$ is formulated as

$$z_t = \sqrt{\bar{\alpha}_t} z_0 + \sqrt{1-\bar{\alpha}_t}\epsilon, \quad \epsilon \sim \mathcal{N}(0, \text{I}) \quad (7)$$

where $\bar{\alpha}_t$ controls the relative proportion of signal and noise. The corresponding SNR is given by

$$\text{SNR}_{\text{diff}}(t) = \frac{\bar{\alpha}_t}{1-\bar{\alpha}_t}. \quad (8)$$

Similarly, the uncertainty-based observation model describes reconstruction difficulty through the predicted variance $z_{\text{var}}$. A larger variance indicates higher uncertainty and thus a lower SNR. Therefore, the uncertainty-based SNR can be written as

$$\text{SNR}_{\text{uncertainty}} = \frac{\sigma_{\text{gt}}^2}{z_{\text{var}}} \quad (9)$$

where $\sigma_{\text{gt}}^2$ denotes the signal variance of the HR latent distribution. By aligning the SNR of the diffusion process with the SNR implied by the uncertainty model, we obtain

$$\bar{\alpha}_{\text{target}} = \frac{\sigma_{\text{gt}}^2}{\sigma_{\text{gt}}^2 + z_{\text{var}}}. \quad (10)$$

The above derivation establishes a theoretical tendency that higher reconstruction uncertainty generally implies a lower SNR and thus a larger timestep, introducing stronger generative guidance. This relationship serves as a design principle rather than a hard constraint, since the optimal denoising timestep also depends on content-dependent signal statistics and structural context. Direct evaluation of (10) is impractical because $\sigma_{\text{gt}}^2$ is unknown and the Gaussian assumption cannot fully characterize the complex distribution of real remote sensing imagery. Therefore, we use (10) to motivate the predictor design and estimate $\bar{\alpha}$target with a data-driven network

$$\bar{\alpha}_{\text{target}} = \text{Pred}_{\theta}\left(z_{\text{gt}}, z_{\text{var}}\right) \quad (11)$$

where $z_{\text{var}}$ provides uncertainty information, while $z_{\text{gt}}$ is available during teacher model construction and provides latent structural reference information. The predictor is optimized jointly with the teacher model through the final reconstruction and adversarial objectives, allowing it to learn the noise level that yields the best restoration performance. It is worth noting that this prediction module is only used for constructing the teacher model and is not involved in the inference of the final efficient student model.

After obtaining $\bar{\alpha}_{\text{target}}$, we need to convert it into the corresponding timestep $t^*$. Since $\bar{\alpha}_t$ is determined by the predefined diffusion noise schedule, this conversion can be written as $t^* = \bar{\alpha}^{-1}(\bar{\alpha}_{\text{target}})$. To avoid the potential artifacts from neural network-based inversion, we adopt a piecewise linear interpolation scheme based on the precomputed noise schedule. Specifically, we maintain two lookup tables: an ascending sequence of noise levels $\{\bar{\alpha}_i\}_{i=0}^{1000}$ sampled from the original diffusion scheduler and the corresponding timestep indices $\{t_i\}_{i=0}^{1000}$. For a given $\bar{\alpha}_{\text{target}}$, we use binary search to locate the nearest pair of entries $(\bar{\alpha}_{j-1}, \bar{\alpha}_j)$ such that $\bar{\alpha}_{j-1} \le \bar{\alpha}_{\text{target}} \le \bar{\alpha}_j$. We then perform linear interpolation in timestep space

$$t^* = t_{j-1} + \frac{\bar{\alpha}_{\text{target}} - \bar{\alpha}_{j-1}}{\bar{\alpha}_j - \bar{\alpha}_{j-1}}\left(t_j - t_{j-1}\right). \quad (12)$$

The resulting timestep inversion strategy converts reconstruction uncertainty into an adaptive diffusion timestep through a differentiable noise-level prediction followed by fast lookup-based inversion.

### C. Efficient Student Model Construction

In the second stage, we aim to lightweight the teacher model obtained in the first stage to derive a more efficient student model, as illustrated in Fig. 3 and Table II. For latent DMs such as SD v2.1, the majority of parameters are concentrated in the UNet, while the variational autoencoder (VAE) contributes a relatively smaller portion of parameters but incurs notable computational overhead because its operations are performed in the HR image space.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 14:26:39 UTC from IEEE Xplore. Restrictions apply.

TABLE II
PROPORTION OF PARAMETERS AND COMPUTATIONAL COMPLEXITY OF VAE AND UNET IN A SINGLE-STEP SD MODEL

| Component | Parameter Ratio | Computation Ratio (MACs) |
|---|---|---|
| UNet | 92.02% | 16% |
| VAE | 7.98% | 84% |

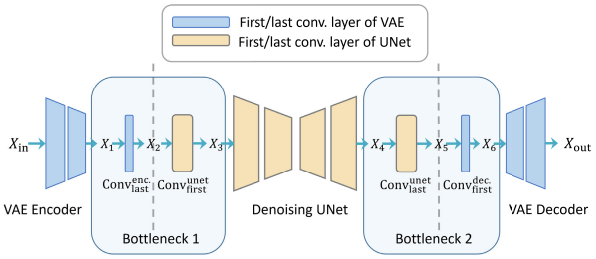



Fig. 5. For the bottleneck between the VAE and the UNet, the last convolutional layer of the VAE encoder and the first convolutional layer of the UNet, respectively, perform dimensionality reduction and expansion on the latent features. These two layers together form a bottleneck structure. During the transformation from $X_1$ to $X_2$, the feature is heavily compressed and cannot be fully recovered in the subsequent mapping from $X_2$ to $X_3$. A similar situation also occurs between the last convolutional layer of the UNet and the first convolutional layer of the VAE decoder.

As summarized in Table II, the UNet overwhelmingly dominates the model capacity, accounting for approximately 92.02% of the total parameters while contributing only 16% of the overall computational cost. In contrast, the VAE contains merely 7.98% of the parameters but incurs a significantly higher computational burden, accounting for about 84% of the total computation. This discrepancy reveals a pronounced imbalance between parameter distribution and computational cost across different components. Therefore, we apply dedicated lightweight optimization strategies to both the VAE and the UNet, leading to the following three approaches.

*1) Enhanced Lightweight VAE:* To reduce the additional computational cost and parameter overhead introduced by the VAE, we replace the original VAE in the latent DM with an extremely lightweight VAE model, TAESD,[1] a compact autoencoder that approximates the functionality of the VAE of SD models while significantly reducing model size and inference cost. Although efficient, TAESD still fails to address the information bottleneck between the UNet and the VAE in the original SD architecture. As illustrated in Fig. 5, the last convolutional layer of the VAE encoder significantly compresses the number of feature channels, while the first convolutional layer of the UNet substantially expands the feature dimensionality

$$X_2 = \mathrm{Conv}^{\mathrm{enc.}}_{\mathrm{last}}(X_1), \quad X_3 = \mathrm{Conv}^{\mathrm{unet}}_{\mathrm{first}}(X_2) \tag{13}$$

where $X_1 \in \mathbb{R}^{B\times 64\times H\times W}$, $X_2 \in \mathbb{R}^{B\times 4\times H\times W}$, and $X_3 \in \mathbb{R}^{B\times 320\times H\times W}$ denote the feature representations before compression, after compression, and after expansion, respectively. Here, the change in the three feature channel dimensions indicates that the feature information is heavily compressed when transforming from $X_1$ to $X_2$, which creates an information bottleneck between the VAE encoder and the UNet. Similarly, the same information bottleneck also exists between the last convolutional layer of the UNet and the first convolutional layer of the VAE decoder.

The presence of such bottleneck layers significantly compresses the edge details of the LR input images, which can adversely affect the subsequent reconstruction process, particularly in terms of fidelity. To address this issue, we propose a convolution fusion strategy, a simple yet effective approach to alleviate the information bottleneck between the VAE and the UNet. Specifically, we fuse the two consecutive convolution operations into a single convolutional transformation. Instead of first compressing the feature channels from 64 to 4 and then expanding them to 320, the proposed strategy directly projects the feature representation from 64 channels to 320 channels

$$X_3 = \mathrm{Conv}_{\mathrm{fuse}}(X_1) \tag{14}$$

where $\mathrm{Conv}_{\mathrm{fuse}}$ denotes the fused convolution operation. By bypassing the intermediate four-channel latent representation $X_2$, the proposed design effectively removes the severe information bottleneck between the VAE encoder and the UNet, allowing richer structural and edge information to be preserved during feature propagation. Similarly, we employ another fused convolution to replace the last convolutional layer of the UNet and the first convolutional layer of the VAE decoder, thereby alleviating the information bottleneck between the UNet and the VAE decoder.

*2) Module Removal in UNet:* The original UNet in SD is designed for generative tasks and thus contains many components that are redundant for SR. We adopt two strategies to remove these unnecessary modules. First, following AdcSR [21], we remove the text encoder and cross-attention (CA) layers, since textual conditioning provides limited guidance for the pruned student network. We also remove time embeddings in the student model. This design is motivated by the different roles of the teacher and student models. The teacher model explicitly uses timestep conditioning to adapt its generative strength according to reconstruction difficulty. We also remove the timestep embedding from the student model. The teacher uses adaptive timesteps to produce high-quality LR-specific SR features, which are directly transferred to the student through distillation. Since the effect of timestep selection is already reflected in the teacher features, the student requires no explicit timestep prediction during inference. Therefore, explicit timestep conditioning becomes redundant for the student model. Removing the timestep embedding thus simplifies the model and improves efficiency, without sacrificing reconstruction quality.

Next, we consider how to further prune the remaining modules. AdcSR applies uniform channel pruning to UNet modules of different depths. However, previous studies [27], [60] have shown that deep modules in the U-Net tend to primarily capture structural and semantic information, whereas shallow modules are more involved in reconstructing fine details and appearance. As shown in Fig. 6, we provide different prompts for the CA layers at different depths. From the final generated images, it is evident that the semantics of the results

[1] https://github.com/madebyollin/taesd

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 14:26:39 UTC from IEEE Xplore. Restrictions apply.

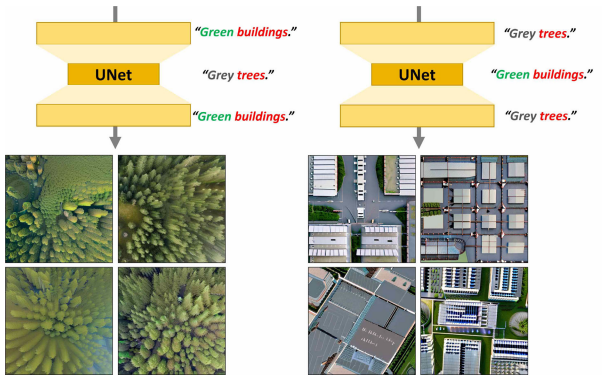



Fig. 6. We provide different textual prompts for the CA layers at varying depths of the UNet. In the figure, we have omitted the prompt "a satellite image with a top-down perspective" and only retained keywords related to style and content, such as "green/gray" and "trees/buildings." It can be observed that the details of the generated image are determined by the shallow layers of the network, while the global content is governed by the deeper layers.

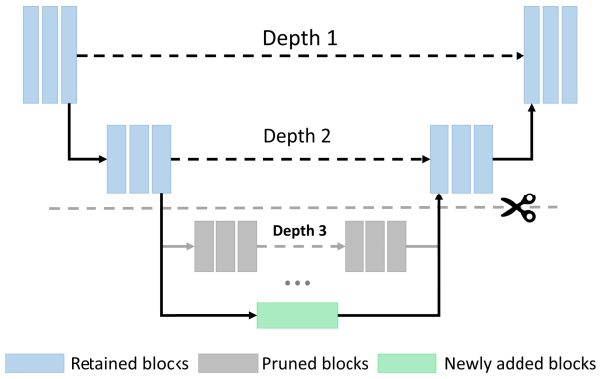



Fig. 7. Our proposed SAP strategy entirely removes the two deepest layers of the UNet, while all shallow layers are fully retained. To ensure structural consistency, the removed deep modules are replaced with a simple ResBlock to preserve the skip connections between the encoder and decoder of the UNet.

follow the deep prompts, while the appearance adheres to the shallow prompts. However, for SR tasks, input LR images already provide sufficient semantic information, making the deep modules in UNet redundant for SR. Therefore, unlike AdcSR, which applies uniform channel pruning across UNet modules at different depths, we propose a semantic-aware pruning (SAP) strategy that completely removes the deeper UNet blocks and replaces them with a single ResBlock to maintain structural integrity, as illustrated in Fig. 7. Since deeper modules typically contain a larger number of channels, pruning them can significantly reduce the model parameters while minimizing the impact on the reconstruction capability of the student model.

*3) Module Replacement in UNet:* To further improve efficiency, we replace the remaining UNet modules with lightweight components tailored to remote sensing imagery. It should be emphasized that these modules are designed primarily as lightweight substitutes for standard operations, rather than as performance-enhancing modules. Their objective is to reduce model size and computational cost while preserving reconstruction capability. Remote sensing images typically

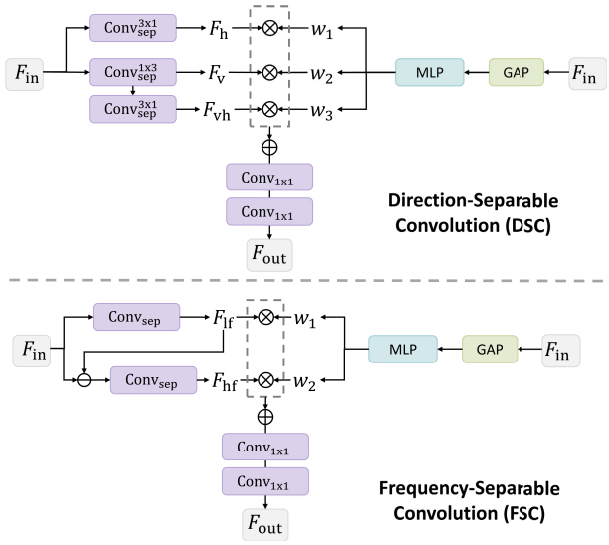



Fig. 8. Architecture diagrams of DSC and FSC. Both modules follow a similar pipeline: they first employ specialized convolutions to extract multisource features, then perform weighted fusion over these features, and finally use two $1 \times 1$ convolutions to complete channel transformation.

exhibit large homogeneous regions with limited low-frequency variation, while critical information is concentrated in sparse high-frequency structures such as edges and object boundaries. To exploit this property, we introduce FSC, which decouples low- and high-frequency components, allowing the network to focus computational resources on informative details. In addition, remote sensing scenes often contain strong directional patterns, such as roads, rivers, and building layouts. To capture such anisotropic structures efficiently, we propose DSC, which decomposes convolution operations into directional components. As illustrated in Fig. 8, both modules follow a similar pipeline: they first employ specialized convolutions to extract features from different aspects, then perform weighted fusion over these features, and finally use two $1 \times 1$ convolutions to complete channel transformation. For DSC, we use horizontal and vertical convolutions to extract features along horizontal, vertical, and diagonal directions. For FSC, we employ two convolutions to separately capture low-frequency information from the input features and high-frequency information from the residual features. It is worth noting that, except for the final two $1 \times 1$ convolutions used for channel transformation, all other convolutions are depthwise separable convolutions, which significantly reduce the number of parameters compared to the standard $3 \times 3$ convolutions in the original UNet.

Furthermore, due to the large spatial extent of remote sensing imagery, long-range dependencies are essential but computationally expensive to model with standard self-attention. Remote sensing imagery often contains a large number of repetitive and limited land-cover categories [61], which can be effectively represented by a set of learnable queries. Therefore, we introduce a QGAM, which approximates global context modeling through learnable queries, reducing complexity from quadratic to linear while preserving global interactions. As illustrated in Fig. 9, QGAM can be formulated as two CA operations. Given input features

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 14:26:39 UTC from IEEE Xplore. Restrictions apply.

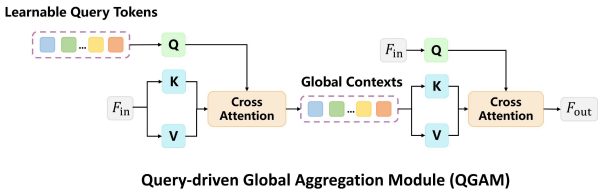



Fig. 9. Architecture diagram of the QGAM. A set of learnable query tokens is first introduced, which interact with the input features via CA to generate global contextual representations. Subsequently, the global context is propagated back to the input features through another CA operation.

$F_{\text{in}} \in \mathbb{R}^{N\times d}$ and learnable queries $Q \in \mathbb{R}^{M\times d}$ $(M \ll N)$, we first aggregate global context tokens

$$G = \text{CA}(Q, F_{\text{in}}) = \text{Softmax}\left(\frac{(QW_q)(F_{\text{in}}W_k)^\top}{\sqrt{d}}\right) F_{\text{in}}W_v \quad (15)$$

and then propagate the global context back to feature tokens

$$F_{\text{out}} = \text{CA}(F_{\text{in}}, G) = \text{Softmax}\left(\frac{(F_{\text{in}}W'_q)(GW'_k)^\top}{\sqrt{d'}}\right) GW'_v. \quad (16)$$

In this way, the computational complexity is reduced from $\mathcal{O}(N^2)$ to $\mathcal{O}(NM)$ while preserving global interactions among input features.

### D. Knowledge Distillation

To minimize the performance gap between the efficient student model and the teacher model, we employ knowledge distillation techniques to transfer the generative capabilities from the teacher to the student. Traditional feature distillation commonly relies on mse loss, which enforces point-wise alignment between corresponding teacher and student features. However, pointwise alignment alone may be insufficient to capture distribution-level discrepancies between feature representations, potentially leading to suboptimal distillation outcomes.

To address this limitation, we introduce the MMD [28] loss in addition to the mse loss to encourage distribution-level alignment between the teacher and student features. MMD is a kernel-based metric that measures the difference between two probability distributions by computing the maximum discrepancy over a class of functions. Specifically, we use the radial basis function (RBF) kernel to compute MMD

$$\text{MMD}^2(P, Q) = \mathbb{E}_{x,x'\sim P}\left[k\left(x, x'\right)\right] + \mathbb{E}_{y,y'\sim Q}\left[k\left(y, y'\right)\right] - 2\mathbb{E}_{x\sim P, y\sim Q}\left[k\left(x, y\right)\right] \quad (17)$$

where $(x, x')$ and $(y, y')$ denote samples drawn from distributions $P$ and $Q$, respectively, and $k(\cdot,\cdot)$ is the RBF kernel defined as $k(x, y) = \exp(-(|x - y|^2)/2\sigma^2)$ with bandwidth $\sigma$.

In our distillation process, for each corresponding layer between the teacher and student networks, we first apply a feature adapter $\phi$ (a lightweight convolutional module) to align the channel dimensions of the student features with those of the teacher. We then compute the mse loss directly on the aligned features

$$\mathcal{L}_{\text{MSE-Dis}} = \frac{1}{N}\sum_{i=1}^{N}\left\|\phi\left(f^i_{\text{stu}}\right) - f^i_{\text{tea}}\right\|^2 \quad (18)$$

where $f^i_{\text{stu}}$ and $f^i_{\text{tea}}$ are the feature maps from the student and teacher networks at layer $i$, and $N$ is the number of remaining UNet layers.

In addition, we flatten the feature maps into sample sets and compute the MMD loss to capture distributional differences

$$\mathcal{L}_{\text{MMD-Dis}} = \text{MMD}^2\left(P_{\text{stu}}, P_{\text{tea}}\right). \quad (19)$$

The total feature distillation loss for each layer is a weighted combination of mse and MMD

$$\mathcal{L}_{\text{Dis.}} = \mathcal{L}_{\text{MSE-Dis}} + \lambda\mathcal{L}_{\text{MMD-Dis}} \quad (20)$$

where $\lambda = 0.1$ is a hyperparameter balancing the two losses. This multiscale feature distillation strategy ensures that the student model not only matches the point-wise statistics but also aligns the distribution-level discrepancies of the intermediate features, leading to improved visual quality and reconstruction fidelity.

### E. Training Scheme

As illustrated in Fig. 3, our method consists of two main training stages: teacher model pretraining and knowledge distillation from teacher to student.

*1) Stage 1 (Teacher Model Pretraining):* In training the teacher model, we freeze the pretrained uncertainty estimator $\text{UE}_\phi$ and use LoRA [62] to fine-tune the linear and convolutional layers in the UNet and VAE-Decoder. For the loss function, we employ a weighted combination of reconstruction and regularization objectives. The reconstruction loss is defined as

$$\mathcal{L}_{\text{Rec.}} = \mathcal{L}_{\text{L1}} + \lambda_{\text{LPIPS}}\mathcal{L}_{\text{LPIPS}} + \lambda_{\text{GAN}}\mathcal{L}_{\text{GAN}} \quad (21)$$

where $\mathcal{L}_{\text{L1}} = \|\hat{x} - x_{\text{gt}}\|_1$, $\mathcal{L}_{\text{LPIPS}}$ is the perceptual loss, and $\mathcal{L}_{\text{GAN}}$ is the adversarial loss. We initialize the discriminator parameters using the pretrained visual encoder DINO [63] to leverage its rich semantic representation capabilities. In addition to reconstruction fidelity, we incorporate a total variation (TV) regularization term on the predicted noise level $\bar{\alpha}_{\text{target}}$ to encourage spatial smoothness

$$\mathcal{L}_{\text{TV}} = \left\|\nabla_h\bar{\alpha}_{\text{target}}\right\|_2^2 + \left\|\nabla_w\bar{\alpha}_{\text{target}}\right\|_2^2 \quad (22)$$

where $\nabla_h$ and $\nabla_w$ denote spatial gradients along height and width dimensions, respectively. This regularization prevents abrupt transitions in the predicted timesteps and promotes consistent adaptive generative guidance across neighboring spatial regions. The total teacher training loss combines both terms

$$\mathcal{L}_{\text{Teacher}} = \mathcal{L}_{\text{Rec.}} + \lambda_{\text{TV}}\mathcal{L}_{\text{TV}}. \quad (23)$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 14:26:39 UTC from IEEE Xplore. Restrictions apply.

*2) Stage 2 (Knowledge Distillation From Teacher to Student):* After completing the teacher model training, we use the teacher model as supervision to train the student model. The retained modules in the student model are initialized with pretrained SD-2.1 parameters, while any newly added modules are initialized with random parameters. All parameters of the student model (including the pruned UNet and enhanced lightweight VAE) participate in training via full fine-tuning. For the loss function, in addition to the same reconstruction losses (21) used in teacher model training, we add the distillation loss (20) to ensure that the generative priors of the teacher model can adequately guide the training of the student model

$$\mathcal{L}_{\text{Student}} = \mathcal{L}_{\text{Rec.}} + \mathcal{L}_{\text{Dis.}}. \tag{24}$$

## IV. Experiments and Discussion

### A. Dataset

We use all 10 000 images from the AID [64] dataset as the training set, and sample 630 images from NWPU-RESISC45 [65], 500 from DOTA [66], and 500 from DIOR [2] as the test set. All images retain their original sizes without any cropping operations. To construct LR images, we follow the degradation process of Real-ESRGAN [67] and consider two magnification factors, 4× and 8×, thereby enabling a comprehensive evaluation of the reconstruction performance of different methods under varying scaling factors.

Beyond synthetic degradation, we also evaluate on a real-world remote sensing dataset, Real-RefRSSRD [68], which provides paired images at different spatial resolutions from actual remote sensing systems. Specifically, the LR images are from Sentinel-2 with a ground sampling distance (GSD) of 10 m, and the HR images are from National Agriculture Imagery Program (NAIP) data with a GSD of 1 m. This dataset is particularly challenging because it involves real degradation processes and large-scale spatial variations inherent to actual Earth observation platforms, rather than synthetic degradation commonly used in standard benchmarks.

### B. Implementation Details

We conduct the two-stage training on 2 RTX 3090 GPUs. In training the teacher model, we use the AdamW optimizer with a learning rate of $2e^{-4}$ for 25 000 iterations with a batch size of 2. The LoRA ranks for the UNet and VAE-Decoder are set to 32 and 16, respectively. During training, input images are randomly cropped to 512 × 512. The weights $\lambda_{\text{LPIPS}}$ and $\lambda_{\text{GAN}}$ are set to 2.5 and 0.25, respectively. In training the student model, we use the Adam optimizer with a learning rate of $5e^{-5}$ for 100 000 iterations with a batch size of 16. The deepest two layers of the student model's UNet are completely removed and replaced with a simple ResBlock. For the remaining UNet modules, we replace them with our proposed lightweight alternative modules. During training, input images are also randomly cropped to 512 × 512.

### C. Evaluation Metrics

To comprehensively evaluate different SR methods, we adopt both full-reference and no-reference image quality metrics. For pixel-level fidelity, we employ PSNR and SSIM, both computed on the Y channel of the YCbCr color space. For perceptual quality assessment, we utilize learned perceptual image patch similarity (LPIPS) and deep image structure and texture similarity (DISTS).

Furthermore, two no-reference metrics, MUSIQ [69] and MANIQA [70], are introduced to provide a more holistic evaluation of overall performance. In terms of efficiency, we consider inference time, model size (number of parameters), and computational complexity (MACs). The inference time is measured in seconds for a 512 × 512 image on a single RTX 3090 GPU. It is worth noting that, due to differences in counting frameworks and implementation details, the reported MACs of our method may differ slightly from the values given in the original papers of the compared methods.

### D. Comparison With State-of-the-Art Methods

To comprehensively evaluate the performance of our proposed method, we compare it against eight state-of-the-art SR methods, including two recent regression-based Transformer methods (TTST [71] and TransENet [72]), two classical GAN-based methods (ESRGAN [73] and SwinIR-GAN [74]), five recent diffusion-based methods (EDiffSR [45], FastDiffSR [44], DiffBIR [17], S3Diff [24], and CTMSR [75]), and one closely related efficient diffusion-based SR method (AdcSR [21]). For a fair comparison, all competing methods are retrained on our dataset using their original code and configurations.

*1) Quantitative Comparison:* Table III presents the quantitative comparison results on the test datasets for 4× SR. Our method achieves competitive performance across various metrics while maintaining high inference efficiency.

First, in fidelity-oriented pixel metrics such as PSNR and SSIM, regression-based methods (TTST and TransENet) achieve the best scores, as they are directly optimized toward pixel-wise losses. However, they tend to produce over-smoothed results, leading to inferior perceptual performance. Second, for perceptual full-reference metrics such as LPIPS and DISTS, our method achieves the best performance. This advantage arises from its ability to leverage the intrinsic priors of pretrained DMs, together with a compact architecture that facilitates efficient adaptation to the distribution of remote sensing imagery. Finally, multistep DMs such as DiffBIR, as well as unpruned diffusion methods like S3Diff, exhibit the strongest generative capacity and therefore achieve superior results on no-reference metrics such as MANIQA and MUSIQ. However, they often introduce hallucinated structures, which degrade performance on full-reference metrics. In particular, the multistep DM FastDiffSR exhibits relatively limited performance on full-reference metrics. Although the single-step consistency model CTMSR achieves competitive PSNR and SSIM among efficient diffusion-based methods, it still underperforms our method on perceptual metrics, including LPIPS and DISTS. Overall, our method not only preserves superior perceptual fidelity but also effectively exploits the generative capability provided by pretrained diffusion priors, achieving a well-balanced reconstruction performance across all metrics.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 14:26:39 UTC from IEEE Xplore. Restrictions apply.

TABLE III

QUANTITATIVE COMPARISON OF DIFFERENT METHODS ON THE 4× TEST DATASETS. THE BEST, SECOND-BEST, AND THIRD-BEST RESULTS ARE HIGHLIGHTED IN BOLD RED, UNDERLINED BLUE, AND ITALIC GREEN, RESPECTIVELY

| Category | Method | Image Quality Metrics | | | | | | Efficiency Metrics | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | PSNR ↑ | SSIM ↑ | LPIPS ↓ | DISTS ↓ | MANIQA ↑ | MUSIQ ↑ | Time (s) ↓ | Param. (M) ↓ | MACs (G) ↓ |
| **Regression-based** | TTST | 26.05 | **0.6663** | 0.5309 | 0.2825 | 0.2397 | 38.55 | 0.218 | *18.37* | 340.03 |
| | TransENet | **26.17** | 0.6651 | 0.5240 | 0.2826 | 0.2381 | 37.50 | *0.077* | 35.09 | **132.51** |
| **GAN-based** | ESRGAN | 24.71 | 0.5919 | 0.3888 | *0.2074* | 0.2442 | 48.18 | 0.044 | 16.72 | 293.71 |
| | SwinIR-GAN | 25.18 | 0.6100 | *0.3753* | 0.2189 | 0.2416 | 45.34 | 0.132 | **11.46** | 214.23 |
| **Diffusion-based** | EDiffSR | 23.19 | 0.4896 | 0.4512 | 0.2799 | 0.1839 | 42.55 | 6.745 | 30.37 | 30911 |
| | FastDiffSR | 25.32 | 0.6203 | 0.4805 | 0.2688 | 0.2480 | 45.68 | 1.500 | 23.80 | 5366.7 |
| | DiffBIR | 21.55 | 0.3986 | 0.4738 | 0.2489 | **0.4523** | **60.86** | 8.637 | 1717 | 47211 |
| | S3Diff | 23.18 | 0.5112 | 0.3511 | 0.1941 | 0.2855 | 51.02 | 0.736 | 1327 | 2776.7 |
| | CTMSR | *25.51* | *0.6253* | 0.4026 | 0.2089 | 0.1781 | 38.96 | 0.126 | 171.5 | 613.01 |
| | AdcSR | 24.64 | 0.5918 | 0.3935 | 0.2256 | 0.2717 | 44.94 | 0.088 | 456.1 | 549.49 |
| | Ours | 24.09 | 0.5642 | **0.3443** | **0.1819** | *0.2837* | *49.84* | **0.037** | 58.05 | *226.72* |




Fig. 10. Visual comparison of 4× SR results of different methods on the DIOR, DOTA, and NWPU-RESISC45 datasets. The results demonstrate that our method is capable of reconstructing more accurate structures and richer edge details.

In terms of inference efficiency, our method demonstrates substantial improvements over other diffusion-based approaches. For example, compared with the multistep DM DiffBIR and the single-step DM S3Diff, our method achieves more than 200× and 20× speedups in inference time, respectively. Compared with the efficient DM AdcSR, our approach further reduces the number of parameters by nearly 10×, thereby significantly lowering the hardware requirements for

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 14:26:39 UTC from IEEE Xplore. Restrictions apply.

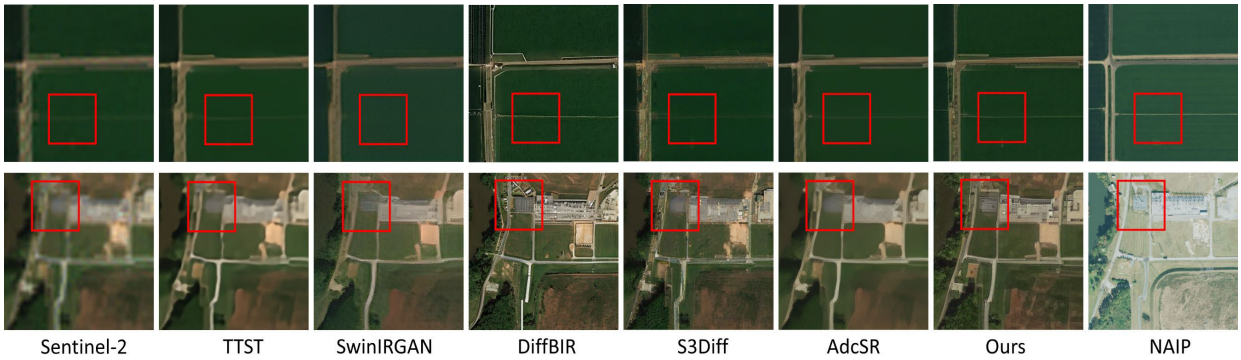



Fig. 11. Visual comparison of 8× SR results produced by different methods on the real-world Sentinel-2 SR task. Due to space limitations, we only present representative diffusion-based and nondiffusion-based methods. Compared with other methods, our approach generates more faithful reconstructions with better preservation of spatial structures and fine details.

deployment. Moreover, it is worth noting that our method even surpasses ESRGAN, a classic GAN-based approach, in inference speed. This is mainly because, benefiting from the unique design of latent DMs, most of the computation in our method is performed in a latent space with an 8× downsampling factor.

*2) Qualitative Comparison:* In Fig. 10, we present a visual comparison of 4× SR results produced by different methods. Also, in Fig. 11, we present a visual comparison on real-world Sentinel-2 SR task. In Fig. 10, the three groups of images, from top to bottom, are sampled from the test sets of the DIOR, DOTA, and NWPU-RESISC45 datasets, respectively. It can be observed that regression-based methods such as TTST and TransENet can only generate overly smooth results, failing to faithfully reconstruct the complex structures and rich edge details in remote sensing imagery. In contrast, ESRGAN and SwinIR-GAN incorporate perceptual losses and adversarial training strategies, leading to significantly improved visual quality compared to the former methods. However, due to the highly complex degradation process in real-world remote sensing images, simple adversarial training is insufficient to introduce strong priors. As a result, although the outputs appear sharper, these methods often produce distorted structures (e.g., the ESRGAN results in the first and second groups). Similarly, diffusion-based methods trained from scratch, including EDiffSR, FastDiffSR, and CTMSR, are unable to exploit the powerful priors learned by large-scale pretrained DMs, and therefore still produce relatively blurry reconstruction results.

By comparison, diffusion-based methods such as DiffBIR and S3Diff effectively leverage pretrained diffusion priors and are able to generate richer details. Nevertheless, they tend to suffer from severe hallucination effects. For instance, in the third group, DiffBIR and S3Diff generate curved and cross-shaped farmland boundaries, respectively, which are inconsistent with real-world land surface structures. As an efficient DM, AdcSR produces overly smooth results. On the one hand, it heavily relies on a pretrained teacher model and is constrained only by an L2 loss in the latent space, which can lead to suboptimal performance when the teacher model is not sufficiently strong. On the other hand, AdcSR adopts uniform channel pruning across all layers of UNet without fully considering the characteristics of the SR task. In contrast, our method, while remaining highly efficient, is still capable of reconstructing competitive structures and fine details, thereby providing a more balanced tradeoff between efficiency and reconstruction quality.

TABLE IV

QUANTITATIVE COMPARISON OF DIFFERENT METHODS ON THE 8× TEST DATASETS. THE BEST, SECOND-BEST, AND THIRD-BEST RESULTS ARE HIGHLIGHTED IN BOLD RED, UNDERLINED BLUE, AND ITALIC GREEN, RESPECTIVELY

| Category | Method | LPIPS ↓ | DISTS ↓ | MANIQA ↑ | MUSIQ ↑ |
|---|---|---|---|---|---|
| **Regression-based** | TTST | 0.6387 | 0.3430 | 0.1968 | 30.74 |
| | TransENet | 0.6240 | 0.3399 | 0.1979 | 31.21 |
| **GAN-based** | ESRGAN | 0.4901 | *0.2715* | 0.2526 | 44.00 |
| | SwinIR-GAN | *0.4865* | 0.2718 | 0.2199 | 39.94 |
| **Diffusion-based** | EDiffSR | 0.5117 | 0.3118 | 0.1842 | 43.47 |
| | FastDiffSR | 0.5424 | 0.2872 | 0.2093 | 39.12 |
| | DiffBIR | 0.5270 | 0.2764 | **0.4524** | **60.34** |
| | S3Diff | **0.3995** | **0.2172** | *0.2832* | *52.79* |
| | CTMSR | 0.4407 | 0.2307 | 0.1675 | 40.00 |
| | AdcSR | 0.5026 | 0.3113 | 0.2535 | 39.42 |
| | Ours | 0.4423 | 0.2461 | 0.3043 | 55.32 |

*3) Results of 8× SR:* Table IV presents the quantitative comparison of different methods on the 8× test set. Since pixel-level metrics (PSNR and SSIM) become less informative at higher scaling factors [76], we primarily adopt four perceptual metrics (LPIPS, DISTS, MANIQA, and MUSIQ) to evaluate reconstruction quality. It can be observed that our method still maintains satisfactory performance under the more challenging 8× SR setting. However, compared with the 4× case, unpruned models such as S3Diff and DiffBIR demonstrate stronger performance than our approach. This is because, under more severe degradation, the pretrained diffusion priors play a more critical role in improving reconstruction quality. In Fig. 12, we present a visual comparison between our method and other representative diffusion-based approaches. The results show that under the challenging 8× SR setting, DiffBIR suffers from severe hallucination artifacts, producing distorted and unrealistic structures. In contrast, S3Diff, as a single-step DM, enables direct supervision in the pixel space, leading to more realistic and faithful details. Similar to the observations in the 4× setting, AdcSR exhibits noticeable blurring in its reconstructions. By comparison, our method achieves reconstruction quality comparable to S3Diff while

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 14:26:39 UTC from IEEE Xplore. Restrictions apply.

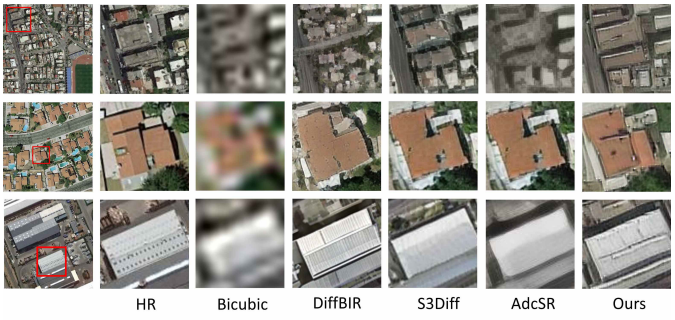



Fig. 12. Visual comparison of 8× SR results of different methods on the DIOR, DOTA, and NWPU-RESISC45 datasets.

TABLE V

QUANTITATIVE COMPARISON OF DIFFERENT METHODS ON THE 8× SENTINEL-2 TEST SET. THE BEST, SECOND-BEST, AND THIRD-BEST RESULTS ARE HIGHLIGHTED IN BOLD RED, UNDERLINED BLUE, AND ITALIC GREEN, RESPECTIVELY

| Category | Method | NIQE↓ | CLIPIQA↑ | MANIQA↑ | MUSIQ↑ |
|---|---|---|---|---|---|
| **Regression-based** | TTST | 10.34 | 0.1929 | 0.2076 | 22.33 |
| | TransENet | 9.442 | 0.1742 | 0.1993 | 22.67 |
| **GAN-based** | ESRGAN | *4.229* | *0.5092* | *0.2428* | 37.84 |
| | SwinIR-GAN | 5.365 | 0.2995 | 0.1940 | 28.62 |
| **Diffusion-based** | EDiffSR | **3.651** | 0.3567 | 0.1840 | 39.69 |
| | FastDiffSR | 7.629 | 0.2463 | 0.2093 | 39.12 |
| | DiffBIR | 4.059 | **0.7010** | **0.4243** | **58.90** |
| | S3Diff | 4.914 | 0.3363 | 0.2425 | 35.57 |
| | CTMSR | 5.577 | 0.4654 | 0.1675 | *40.00* |
| | AdcSR | 6.540 | 0.2652 | 0.1871 | 33.12 |
| | Ours | 5.172 | 0.5100 | 0.2579 | 46.95 |

significantly improving inference efficiency, demonstrating both its effectiveness and efficiency.

*4) Results of Real-World Dataset:* To further validate effectiveness in real-world scenarios, we evaluate SlimDiffSR on the Real-RefRSSRD dataset [68]. It is important to note that we do not use the cross-sensor paired data provided by Real-RefRSSRD for training. Therefore, the model is not expected to generate SR outputs with NAIP-specific radiometric characteristics. NAIP images are used only as visual References rather than strict radiometric supervision targets.

Therefore, due to the lack of ground-truth data, Table V presents the quantitative comparison using nonreference metrics, including NIQE, CLIPIQA, MANIQA, and MUSIQ. DiffBIR achieves the best performance by leveraging multistep denoising and strong priors. Our method achieves second-best results on most metrics while maintaining significant efficiency advantages. Qualitatively, as shown in Fig. 11, our efficient model balances reconstruction fidelity and efficiency effectively. While DiffBIR produces visually sharp results, it suffers from hallucination artifacts. In contrast, our method generates faithful spatial structures, preserved boundaries, and rich details comparable to S3Diff while achieving substantially faster inference. These results indicate that the proposed efficient student is not merely overfitted to the synthetic degradation pipeline, but can generalize to real-world remote sensing degradations while maintaining stable structural reconstruction.

*5) Performance on Downstream Tasks:* To further evaluate the practical utility of reconstructed images, we conduct

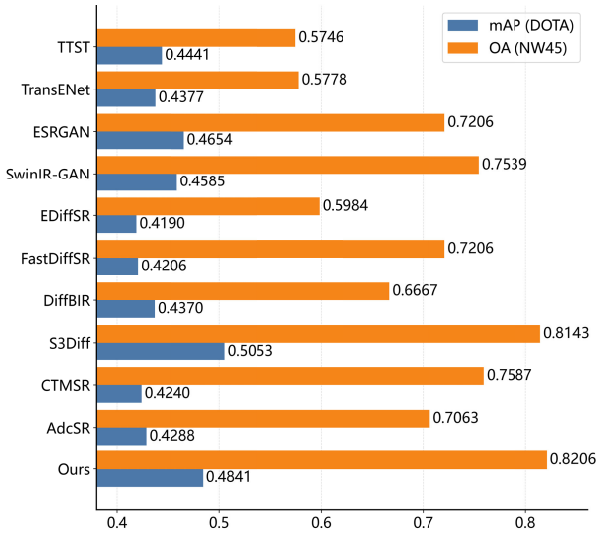



Fig. 13. Downstream task evaluation on SR outputs. Rotated object detection is performed using pretrained RTMDet-L on DOTA (mAP), and scene classification is performed using pretrained RSMamba on NWPU-RESISC45 (OA). Our method achieves the best OA and competitive mAP, demonstrating strong transferability of reconstructed details and semantics to practical remote sensing tasks.

two representative downstream tasks, including rotated object detection and scene classification. For object detection, we use a pretrained RTMDet-L detector [77] and evaluate on DOTA using mean average precision (mAP). For classification, we use a pretrained RSMamba classifier [78] and evaluate on NWPU-RESISC45 using overall accuracy (OA). Following a common downstream evaluation protocol, we directly feed the SR outputs of each method into these pretrained models without additional fine-tuning, so that the comparison mainly reflects the transferability of restored structures and semantics.

As shown in Fig. 13, our method achieves the best OA (0.8206), indicating that it preserves scene-level semantic consistency more effectively than competing methods. In rotated object detection, our method obtains a competitive mAP of 0.4841, ranking second after S3Diff (0.5053), while clearly outperforming other efficient diffusion baselines such as FastDiffSR (0.4206), CTMSR (0.4240), and AdcSR (0.4288). These results suggest that SlimDiffSR not only improves perceptual reconstruction quality, but also provides more task-relevant features for downstream remote sensing interpretation.

It is worth noting that regression-based methods such as TTST and TransENet usually perform poorly on downstream interpretation tasks. Although they often achieve relatively high PSNR and SSIM values, their reconstructed images are over-smoothed and lack fine structural details, which makes it difficult for downstream algorithms to effectively extract discriminative information from the restored images.

## E. Ablation Studies

In this section, we conduct ablation studies to validate the effectiveness of our design choices for both the teacher and student models. It is worth noting that all metrics are computed on the DIOR test set.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 14:26:39 UTC from IEEE Xplore. Restrictions apply.

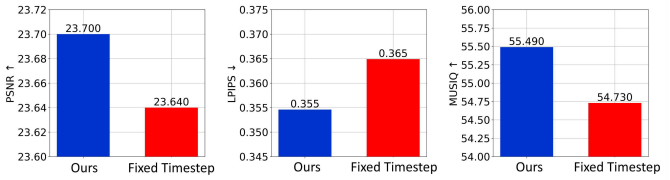



Fig. 14. Ablation analysis of the uncertainty-guided strategy. For comparison, we retrain a simple single-step DM with a fixed time step as the teacher model. The results show that the proposed teacher model achieves consistent improvements across all evaluation metrics.

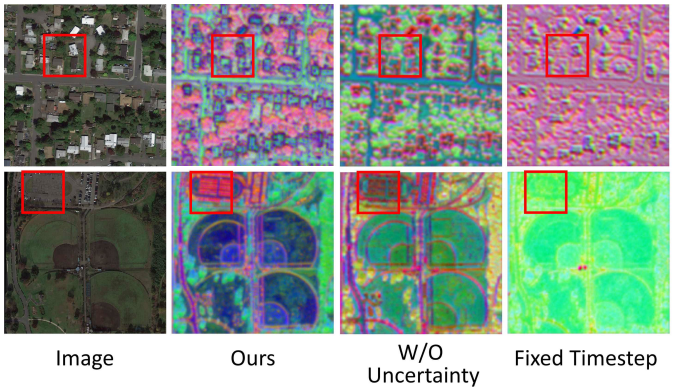



Fig. 15. Visualization of intermediate features from the teacher U-Net. The proposed strategy that jointly combines uncertainty estimation with dynamic timestep assignment encourages the activation of higher quality intermediate features, thereby providing more effective supervision for the student model.

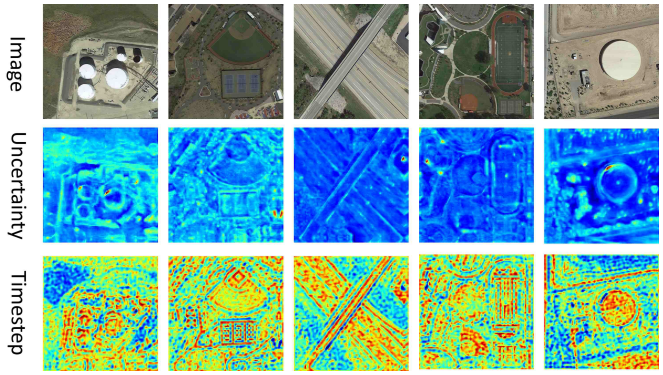



Fig. 16. Visualization of uncertainty maps and the corresponding predicted timestep maps. The uncertainty map reflects reconstruction ambiguity, while the predicted timestep represents the learned adaptive guidance level of the teacher model. Although high-uncertainty regions generally tend to require stronger generative guidance, the timestep map is not a direct transformation of uncertainty, since it is jointly determined by uncertainty and latent structural context.

TABLE VI

ABLATION STUDY ON ENHANCED LIGHTWEIGHT VAE. THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN BOLD RED AND UNDERLINED BLUE, RESPECTIVELY

| Method | Image Quality Metrics | | | Efficiency Metrics | |
|---|---|---|---|---|---|
| | PSNR ↑ | LPIPS ↓ | MUSIQ ↑ | Time (s) ↓ | Params (M) ↓ |
| VAE | **23.95** | **0.3505** | 54.21 | 0.195 | 138.9 |
| TAESD | 23.32 | 0.3621 | 53.87 | **0.037** | **57.68** |
| Ours | 23.70 | 0.3546 | **55.49** | **0.037** | 58.05 |

*1) Effectiveness of Uncertainty Guided Adaptive Timestep:* For the teacher model, we jointly leverage uncertainty information and latent structural information to estimate an adaptive timestep, which is then used to guide the single-step denoising process of the teacher UNet. To validate the effectiveness of this mechanism, we alternatively employ a single-step DM with a fixed timestep as the teacher model and use it to supervise the student model via knowledge distillation. As shown in Fig. 14, the proposed uncertainty-guided adaptive timestep mechanism consistently improves the performance of the student model. This is because, compared with a fixed timestep, the proposed teacher model can dynamically adjust the generative strength of the single-step DM according to the spatial reconstruction difficulty of different inputs.

In Fig. 15, we further visualize the intermediate features from the teacher UNet. First, we remove the uncertainty map and estimate the adaptive timestep using only the $z_{\text{gt}}$ term in (11). Second, we directly employ a fixed timestep for denoising. It can be observed that, compared with the two ablation variants, the proposed uncertainty-guided teacher model is able to activate richer structural features, thereby providing higher quality feature supervision for training the student model in the second stage.

In Fig. 16, we present the uncertainty maps predicted by $\text{UE}_\phi$. Regions with dense structures and rich textures usually exhibit higher uncertainty, indicating greater reconstruction ambiguity. We further visualize the predicted timestep maps. The results show that the timestep prediction is not always strictly monotonic with the uncertainty map. This is reasonable because the predictor is optimized by the final teacher restoration objective and also uses latent structural information during teacher construction. Therefore, in some low-uncertainty regions, larger timesteps may still be selected when stronger diffusion priors help improve structural consistency or perceptual restoration.

*2) Effectiveness of Enhanced Lightweight VAE:* To evaluate the effectiveness of our enhanced lightweight VAE, we compare it against the original large VAE and an extremely lightweight VAE variant (TAESD). The original VAE provides high reconstruction quality but is computationally expensive due to its large model size. TAESD is a lightweight alternative that significantly reduces parameters and inference time but may compromise reconstruction quality due to information bottlenecks in the latent space. Our enhanced VAE addresses this by fusing convolution layers between the VAE and UNet, alleviating the information bottleneck and preserving richer structural details while maintaining efficiency.

Table VI presents the SR performance of the student model equipped with different variants of the VAE. Our method achieves the best MUSIQ score, indicating superior perceptual quality, while maintaining low inference time and moderate parameter count. Compared to the original VAE, it significantly reduces inference time by approximately 5 times and parameters by about 2.4 times, with only a slight drop in PSNR and LPIPS. Compared to TAESD, it improves PSNR by 0.38 and MUSIQ by 1.62, while keeping similar efficiency. This demonstrates that our enhanced VAE effectively balances reconstruction quality and computational efficiency.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 14:26:39 UTC from IEEE Xplore. Restrictions apply.

TABLE VII

ABLATION STUDIES ON MODULE REMOVAL AND MODULE REPLACEMENT STRATEGIES. THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN BOLD RED AND UNDERLINED BLUE, RESPECTIVELY

| Method | Image Quality Metrics | | | Efficiency Metrics | | |
|---|---|---|---|---|---|---|
| | PSNR ↑ | LPIPS ↓ | MUSIQ ↑ | Time (s) ↓ | Params (M) ↓ | MACs (G) ↓ |
| W/O SAP | **23.89** | 0.3667 | 54.96 | 0.052 | 168.3 | 423.81 |
| W/O DSC | 23.67 | 0.3621 | 55.28 | 0.040 | 93.18 | 280.22 |
| W/O FSC | 23.44 | 0.3540 | 55.44 | 0.042 | 99.46 | 295.97 |
| W/O QGAM | 23.34 | **0.3526** | **55.57** | 0.057 | 81.11 | 244.45 |
| Ours | 23.70 | 0.3546 | 55.49 | **0.037** | **58.05** | **226.72** |

*3) Effectiveness of Module Removal Strategy:* To evaluate the effectiveness of our SAP strategy, we compare it against the uniform channel pruning method adopted in AdcSR. As shown in Table VII, our SAP strategy achieves superior performance by semantically identifying and removing redundant modules tailored to the SR task. The results demonstrate that simply removing channels uniformly (W/O SAP) leads to significantly higher parameter counts (168.3 M) and inference time (0.052 s), while substantially degrading perceptual quality (54.96 on MUSIQ). By contrast, our semantic-aware approach effectively reduces the model size while maintaining competitive perceptual quality, highlighting the importance of task-specific pruning strategies for remote sensing image SR.

*4) Effectiveness of Module Replacement Strategy:* Beyond module removal, we further enhance efficiency through replacing standard convolutional and self-attention layers with our proposed lightweight alternatives. The purpose of these modules is not to surpass the original heavy modules in reconstruction accuracy, but to serve as lightweight substitutes that retain comparable SR performance with substantially lower model complexity. The ablation results in Table VII progressively demonstrate the impact of each design choice. When removing the DSC, PSNR drops to 23.67, and the model requires 93.18-M parameters with 0.040-s inference time. The FSC contributes significantly to maintaining fine details, as removing it yields a PSNR of 23.44 and slightly increases inference time to 0.042 s. The QGAM provides crucial global context, and its absence results in the lowest PSNR (23.34) and highest inference time (0.057 s) among the ablated variants. The MACs measurements further quantify this efficiency–performance tradeoff: removing DSC, FSC, or QGAM increases the computation from 226.72 to 280.22, 295.97, and 244.45 G, respectively, while replacing SAP with uniform pruning incurs 423.81-G MACs. Thus, each proposed lightweight design contributes to the compactness of the final model. We observe that incorporating QGAM leads to slight decreases in LPIPS and MUSIQ. This behavior can be attributed to the global smoothing effect of the proposed global aggregation mechanism. Specifically, QGAM may reduce the diversity of high-frequency details, leading to less pronounced texture variations. Overall, these results demonstrate that the proposed module replacement strategy achieves a favorable tradeoff between efficiency and reconstruction quality. This validates that RS-aware lightweight design can serve as an effective alternative to standard modules in diffusion-based SR models without incurring significant performance loss.

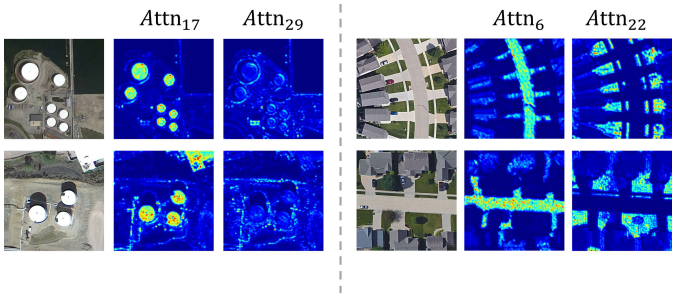



Fig. 17. Visualization of the attention scores in the proposed QGAM module. $Attn_N$ denotes the attention map corresponding to the $N$th learnable query token. It can be observed that, across different scenes, the learnable queries are able to aggregate semantically rich information and respond to specific land-cover types, such as storage tanks, background regions, roads, and grassland. Therefore, this design enables global token interaction with linear computational complexity.

TABLE VIII

ABLATION STUDY ON KNOWLEDGE DISTILLATION LOSSES. THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN BOLD RED AND UNDERLINED BLUE, RESPECTIVELY

| Method | PSNR ↑ | LPIPS ↓ | MUSIQ ↑ |
|---|---|---|---|
| No-Dis | 22.69 | 0.4370 | **57.66** |
| No-MMD | 23.47 | 0.3574 | 53.99 |
| Ours | **23.70** | **0.3546** | 55.49 |

The proposed QGAM enables global token interaction through two CA operations, effectively alleviating the high computational cost of global self-attention. Since remote sensing images are typically composed of a limited set of land-cover categories, each learnable query token in QGAM can act as a semantic proxy to extract specific types of information from the image. In Fig. 17, we visualize the attention maps corresponding to different learnable query tokens. It can be observed that, across various scenes, these queries are able to aggregate rich semantic information and respond to specific land-cover types, such as storage tanks, background regions, roads, and grassland. Therefore, QGAM serves as an efficient alternative to computationally expensive self-attention modules while maintaining inference efficiency.

*5) Effectiveness of Knowledge Distillation:* To verify the contribution of the distillation objective, we compare three variants: No-Dis (without any distillation loss), No-MMD (using only mse distillation), and Ours (using both mse and MMD losses). As shown in Table VIII, introducing distillation significantly improves reconstruction fidelity. Compared with No-Dis, the No-MMD variant improves PSNR from 22.69 to 23.47 and reduces LPIPS from 0.4370 to 0.3574, demonstrating that feature-level supervision from the teacher model is essential for preserving faithful structures and textures. It is worth noting that the No-Dis variant achieves a relatively high MUSIQ score. This is because, without any feature-level constraints, the model tends to generate excessive textures. However, these textures are often unrealistic and inconsistent with the ground truth, which leads to a significant degradation in full-reference metrics such as PSNR and LPIPS.

Furthermore, adding the proposed MMD loss on top of mse leads to consistent gains across all metrics. Specifically, our full method further improves PSNR to 23.70, reduces LPIPS

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 14:26:39 UTC from IEEE Xplore. Restrictions apply.




Fig. 18. Visual comparison for the ablation study on knowledge distillation. The inclusion of the distillation loss and the MMD loss significantly improves the semantic fidelity of the reconstructed results.

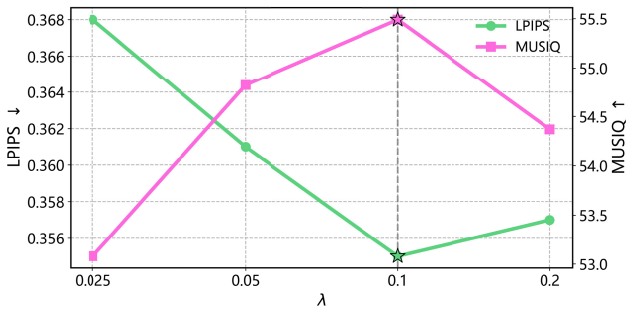



Fig. 19. Ablation study on the effect of the $\lambda$ parameter in the knowledge distillation loss.

to 0.3546, and increases MUSIQ to 55.49 compared with No-MMD. These results indicate that MSE-based pointwise alignment alone is insufficient to fully transfer teacher knowledge, while the additional MMD constraint effectively reduces the distribution gap between teacher and student features. Consequently, the student model can achieve a better balance between perceptual quality and reconstruction fidelity without introducing additional inference overhead. In Fig. 18, we present a visual comparison of reconstruction results from different ablation variants. As observed in the first two rows, the introduction of the distillation loss and the MMD loss significantly improves the semantic consistency of the reconstruction results, as evidenced by clearer building boundaries. From the last row, it can be seen that our method effectively reduces artifacts in the SR images. These results further demonstrate the effectiveness of the proposed knowledge distillation strategy.

In the knowledge distillation loss, the weight $\lambda$ controls the balance between the MMD-based distillation loss and the MSE-based distillation loss. We conduct a sensitivity analysis of $\lambda$, and the results are shown in Fig. 19. It can be observed that both major perceptual metrics achieve the best performance when $\lambda = 0.1$. This indicates that, with this setting, the distillation loss achieves an optimal balance between distribution-level feature alignment and point-wise feature alignment.

### F. Limitation

Although this article presents an efficient diffusion framework for real-world remote sensing image SR, its performance still needs to be improved under more challenging settings such as $8\times$ SR and real-world SR. In addition, the current model is designed for fixed-scale SR and cannot handle arbitrary-scale SR, which is of great significance in remote sensing applications. Therefore, future research could focus on leveraging stronger generative diffusion priors, such as pre-trained diffusion transformer (DiT) priors [79], and exploring more powerful degradation-aware priors to enable arbitrary-scale SR and further improve robustness in complex remote sensing scenarios.

## V. Conclusion

In this article, we propose SlimDiffSR, an efficient diffusion framework for real-world remote sensing image SR. The method follows a two-stage paradigm, where an uncertainty-guided single-step teacher model is first constructed, followed by a compact student model obtained via structured pruning and knowledge distillation. Specifically, we introduce dynamic timestep assignment through uncertainty estimation and timestep inversion, develop an enhanced lightweight VAE and a semantic-aware UNet pruning strategy, and incorporate direction- and frequency-separable operators together with a QGAM to improve efficiency. Moreover, we combine mse and MMD losses to enhance distribution-level feature alignment during distillation, enabling the student model to better approximate the teacher.

Extensive experiments on multiple remote sensing benchmarks demonstrate that SlimDiffSR achieves a favorable balance between reconstruction quality and efficiency, delivering strong perceptual performance with significantly reduced inference cost compared to existing diffusion-based methods. Ablation studies further validate the effectiveness of each component. While the method remains competitive under more challenging settings such as $8\times$ SR and real-world SR, improving robustness under severe degradations and cross-sensor domain gaps remains an important direction. Future work will explore stronger degradation-aware priors, arbitrary-scale SR, and more general efficient diffusion frameworks for large-scale Earth observation applications.

## Acknowledgment

The numerical calculations in this article have been done on the supercomputing system in the Supercomputing Center of Wuhan University.

## References

[1] Z. S. Venter, O. Brousse, I. Esau, and F. Meier, "Hyperlocal mapping of urban air temperature using remote sensing and crowdsourced weather data," *Remote Sens. Environ.*, vol. 242, Jun. 2020, Art. no. 111791.

[2] K. Li, G. Wan, G. Cheng, L. Meng, and J. Han, "Object detection in optical remote sensing images: A survey and a new benchmark," *ISPRS J. Photogramm. Remote Sens.*, vol. 159, pp. 296–307, Jan. 2020.

[3] X.-Y. Tong et al., "Land-cover classification with high-resolution remote sensing images using transferable deep models," *Remote Sens. Environ.*, vol. 237, Feb. 2020, Art. no. 111322.

[4] B. Huang, H. Song, H. Cui, J. Peng, and Z. Xu, "Spatial and spectral image fusion using sparse matrix factorization," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 3, pp. 1693–1704, Mar. 2014.

[5] Z. Wang, J. Chen, and S. C. H. Hoi, "Deep learning for image super-resolution: A survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 10, pp. 3365–3387, Oct. 2021.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 14:26:39 UTC from IEEE Xplore. Restrictions apply.

[6] Y. Zhang, Y. Tian, Y. Kong, B. Zhong, and Y. Fu, "Residual dense network for image super-resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 2472–2481.

[7] J. Kim, J. K. Lee, and K. M. Lee, "Accurate image super-resolution using very deep convolutional networks," in *Proc. IEEE CVPR*, Jun. 2016, pp. 1646–1654.

[8] X. Chen, X. Wang, J. Zhou, Y. Qiao, and C. Dong, "Activating more pixels in image super-resolution transformer," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 22367–22377.

[9] J. Johnson, A. Alahi, and L. Fei-Fei, "Perceptual losses for real-time style transfer and super-resolution," in *Proc. Eur. Conf. Comput. Vis.*, Amsterdam, Netherlands, 2016, pp. 694–711.

[10] C. Ledig et al., "Photo-realistic single image super-resolution using a generative adversarial network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 4681–4690.

[11] I. Goodfellow et al., "Generative adversarial networks," *Commun. ACM*, vol. 63, no. 11, pp. 139–144, 2020.

[12] W. Zhang et al., "Real-world image super-resolution as multi-task learning," in *Proc. NeurIPS*, 2023, pp. 21003–21022.

[13] J. Liang, H. Zeng, and L. Zhang, "Details or artifacts: A locally discriminative learning approach to realistic image super-resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 5657–5666.

[14] J. Ho, "Denoising diffusion probabilistic models," in *Proc. NeurIPS*, vol. 33, 2024, pp. 6840–6851.

[15] Z. Yue, J. Wang, and C. C. Loy, "ResShift: Efficient diffusion model for image super-resolution by residual shifting," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 36, 2023, pp. 13294–13307.

[16] J. Wang, Z. Yue, S. Zhou, K. C. K. Chan, and C. C. Loy, "Exploiting diffusion prior for real-world image super-resolution," *Int. J. Comput. Vis.*, vol. 132, no. 12, pp. 5929–5949, Dec. 2024.

[17] X. Lin et al., "DiffBIR: Toward blind image restoration with generative diffusion prior," in *Proc. ECCV*, 2024, pp. 430–448.

[18] J. Liu, Q. Wang, H. Fan, Y. Wang, Y. Tang, and L. Qu, "Residual denoising diffusion models," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 2773–2783.

[19] J. Song, C. Meng, and S. Ermon, "Denoising diffusion implicit models," in *Proc. ICLR*, 2020, pp. 1–20.

[20] C. Lü, Y. Zhou, F. Bao, J. F. Chen, C. Li, and J. Zhu, "DPM-Solver: A fast ODE solver for diffusion probabilistic model sampling in around 10 steps," in *Proc. NeurIPS*, 2022, pp. 5775–5787.

[21] B. Chen et al., "Adversarial diffusion compression for real-world image super-resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2025, pp. 28208–28220.

[22] B.-K. Kim, H.-K. Song, T. Castells, and S. Choi, "BK-SDM: A lightweight, fast, and cheap version of stable diffusion," in *Proc. ECCV*, 2024, pp. 381–399.

[23] R. Wu, L. Sun, Z. Ma, and L. Zhang, "One-step effective diffusion network for real-world image super-resolution," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 37, 2024, pp. 92529–92553.

[24] A. Zhang, Z. Yue, R. Pei, W. Ren, and X. Cao, "Degradation-guided one-step image super-resolution with diffusion priors," 2024, *arXiv:2409.17058*.

[25] C. Wang, Z. Hu, W. Sun, and Z. Chen, "Timestep-aware diffusion model for extreme image rescaling," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2025, pp. 15594–15603.

[26] N. Qian, W. Dong, X. Li, J. Wu, and G. Shi, "Uncertainty-driven loss for single image super-resolution," in *Proc. NeurIPS*, vol. 34, 2021, pp. 16398–16409.

[27] A. Voynov, Q. Chu, D. Cohen-Or, and K. Aberman, "P+: Extended textual conditioning in text-to-image generation," 2023, *arXiv:2303.09522*.

[28] A. Gretton, K. M. Borgwardt, M. J. Rasch, B. Schölkopf, and A. Smola, "A kernel two-sample test," *J. Mach. Learn. Res.*, vol. 13, no. 1, pp. 723–773, 2012.

[29] C. Dong, C. C. Loy, K. He, and X. Tang, "Learning a deep convolutional network for image super-resolution," in *Proc. Eur. Conf. Comput. Vis. Cham*, Switzerland: Springer, 2014, pp. 184–199.

[30] C. Tuna, G. Unal, and E. Sertel, "Single-frame super resolution of remote-sensing images by convolutional neural networks," *Int. J. Remote Sens.*, vol. 39, no. 8, pp. 2463–2479, Apr. 2018.

[31] N. Granger and M. A. El-Yacoubi, "Comparing hybrid NN-HMM and RNN for temporal modeling in gesture recognition," in *Proc. ICONIP*, 2017, pp. 147–156.

[32] W. Xu, G. Xu, Y. Wang, X. Sun, D. Lin, and Y. Wu, "High Quality remote sensing image super-resolution using deep memory connected network," in *Proc. IEEE IGARSS*, Jul. 2018, pp. 8889–8892.

[33] C. Lanaras, J. Bioucas-Dias, S. Galliani, E. Baltsavias, and K. Schindler, "Super-resolution of sentinel-2 images: Learning a globally applicable deep neural network," *ISPRS J. Photogramm. Remote Sens.*, vol. 146, pp. 305–319, Dec. 2018.

[34] Y.-Z. Chen, T.-J. Liu, and K.-H. Liu, "Super-resolution of satellite images based on two-dimensional RRDB and edge-enhanced generative adversarial network," in *Proc. IEEE Int. Conf. Consum. Electron. (ICCE)*, Jan. 2022, pp. 1–4.

[35] W. Ma, Z. Pan, F. Yuan, and B. Lei, "Super-resolution of remote sensing images via a dense residual generative adversarial network," *Remote Sens.*, vol. 11, no. 21, p. 2578, Nov. 2019.

[36] F. Meng et al., "Single remote sensing image super-resolution via a generative adversarial network with stratified dense sampling and chain training," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5400822.

[37] Y. Li et al., "Single-image super-resolution for remote sensing images using a deep generative adversarial network with local and global attention mechanisms," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–24, 2022.

[38] B. Liu, H. Li, Y. Zhou, Y. Peng, A. Elazab, and C. Wang, "A super resolution method for remote sensing images based on cascaded conditional Wasserstein GANs," in *Proc. IEEE ICICSP*, Sep. 2020, pp. 284–289.

[39] J. Zhang, T. Xu, J. Li, S. Jiang, and Y. Zhang, "Single-image super resolution of remote sensing images with real-world degradation modeling," *Remote Sens.*, vol. 14, no. 12, p. 2895, Jun. 2022.

[40] C. Saharia, J. Ho, W. Chan, T. Salimans, D. J. Fleet, and M. Norouzi, "Image super-resolution via iterative refinement," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 4, pp. 4713–4726, Apr. 2023.

[41] H. Li et al., "SRDiff: Single image super-resolution with diffusion probabilistic models," *Neurocomputing*, vol. 479, pp. 47–59, Mar. 2022.

[42] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer, "High-resolution image synthesis with latent diffusion models," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 10674–10685.

[43] X. Li et al., "LinearSR: Unlocking linear attention for stable and efficient image super-resolution," in *Proc. ICLR*, 2025, pp. 1–24.

[44] F. Meng et al., "A conditional diffusion model with fast sampling strategy for remote sensing image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5408616.

[45] Y. Xiao, Q. Yuan, K. Jiang, J. He, X. Jin, and L. Zhang, "EDiffSR: An efficient diffusion probabilistic model for remote sensing image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5601514.

[46] F. Meng et al., "Remote sensing image blind super-resolution via a convolutional neural network-guided conditional diffusion model," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 19, pp. 2641–2657, 2026.

[47] F. Meng, Z. Jiang, F. Xie, S. Wu, Z. Du, and H. Zhang, "Single remote sensing image super-resolution via convolutional neural network and diffusion model," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 19, pp. 16825–16840, 2026.

[48] C. Wang and W. Sun, "Semantic guided large scale factor remote sensing image super-resolution with generative diffusion prior," *ISPRS J. Photogramm. Remote Sens.*, vol. 220, pp. 125–138, Feb. 2025.

[49] S. Khanna et al., "DiffusionSat: A generative foundation model for satellite imagery," in *Proc. ICLR*, 2023, pp. 1–19.

[50] H. Wu et al., "CT-HiffNet: A contour-texture hierarchical feature fusion network for cropland field parcel extraction from high-resolution remote sensing images," *Comput. Electron. Agricult.*, vol. 239, Dec. 2025, Art. no. 111010.

[51] V. Spasev, I. Dimitrovski, I. Chorbev, and I. Kitanovski, "Semantic segmentation of unmanned aerial vehicle remote sensing images using SegFormer," in *Proc. Int. Conf. Intell. Syst. Pattern Recognit.*, 2024, pp. 108–122.

[52] M. Liu and H. Yin, "Efficient pyramid context encoding and feature embedding for semantic segmentation," *Image Vis. Comput.*, vol. 111, Jul. 2021, Art. no. 104195.

[53] J. Xie et al., "A CNN-transformer hybrid network with boundary guidance for mapping cropland field parcels from high-resolution remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, 2026, Art. no. 4401122.

[54] X. Junyang, L. Anqi, W. Hao, W. Ziwei, W. Wenbin, and Y. Qiangyi, "JAM-R-CNN deep learning network model for remote sensing recognition of terraced fields," *Nat. Remote Sens. Bull.*, vol. 28, no. 12, pp. 3136–3146, 2024.

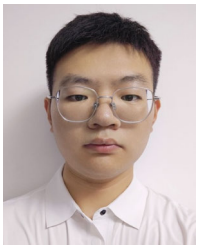

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 14:26:39 UTC from IEEE Xplore. Restrictions apply.

[55] J. Xie et al., "S2-IFNet: A spatial-semantic information fusion network integrated with boundary feature enhancement for forest land extraction from sentinel-2 data," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 139, May 2025, Art. no. 104505.
[56] L. Zhou, T. Peng, Z. Han, L. Li, Q. Zhu, and Y. Ye, "Robust pixel-by-pixel multimodal remote sensing image registration using geometry preserving dense registration network," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, 2026, Art. no. 4701814.
[57] X. Li, C. Li, P. Ghamisi, D. Hong, J. A. Benediktsson, and J. Chanussot, "FlexiMo: A flexible remote sensing foundation model," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, 2026, Art. no. 5606516.
[58] X. Kong, H. Zhao, Y. Qiao, and C. Dong, "ClassSR: A general framework to accelerate super-resolution networks by data characteristic," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2021, pp. 12016–12025.
[59] M. A. T. Figueiredo, "Adaptive sparseness using Jeffreys' prior," in *Proc. NeurIPS*, 2002, pp. 697–704.
[60] X. Li et al., "MegaSR: Mining customized semantics and expressive guidance for real-world image super-resolution," 2025, *arXiv:2503.08096*.
[61] X. Guo et al., "SkySense: A multi-modal remote sensing foundation model towards universal interpretation for Earth observation imagery," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 27662–27673.
[62] E. J. Hu et al., "LoRA: Low-rank adaptation of large language models," in *Proc. ICLR*, 2022, vol. 1, no. 2, p. 3.
[63] M. Caron et al., "Emerging properties in self-supervised vision transformers," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 9630–9640.
[64] G.-S. Xia et al., "AID: A benchmark data set for performance evaluation of aerial scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3965–3981, Jul. 2017.
[65] G. Cheng, J. Han, and X. Lu, "Remote sensing image scene classification: Benchmark and state of the art," *Proc. IEEE*, vol. 105, no. 10, pp. 1865–1883, Oct. 2017.
[66] G.-S. Xia et al., "DOTA: A large-scale dataset for object detection in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 3974–3983.
[67] X. Wang, L. Xie, C. Dong, and Y. Shan, "Real-ESRGAN: Training real-world blind super-resolution with pure synthetic data," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshops (ICCVW)*, Oct. 2021, pp. 1905–1914.
[68] C. Wang and W. Sun, "Controllable reference-guided diffusion with local–global fusion for real-world remote sensing super resolution," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 19, pp. 4963–4979, 2026.
[69] J. Ke, Q. Wang, Y. Wang, P. Milanfar, and F. Yang, "MUSIQ: Multi-scale image Quality transformer," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 5148–5157.
[70] S. Yang et al., "MANIQA: Multi-dimension attention network for no-reference image Quality assessment," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2022, pp. 1191–1200.
[71] Y. Xiao, Q. Yuan, K. Jiang, J. He, C.-W. Lin, and L. Zhang, "TTST: A top-k token selective transformer for remote sensing image super-resolution," *IEEE Trans. Image Process.*, vol. 33, pp. 738–752, 2024.
[72] S. Lei, Z. Shi, and W. Mo, "Transformer-based multistage enhancement for remote sensing image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–11, 2022.
[73] X. Wang et al., "ESRGAN: Enhanced super-resolution generative adversarial networks," in *Proc. ECCV Workshops*, 2018, pp. 63–79.
[74] J. Liang, J. Cao, G. Sun, K. Zhang, L. Van Gool, and R. Timofte, "SwinIR: Image restoration using Swin transformer," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshops (ICCVW)*, Oct. 2021, pp. 1833–1844.
[75] W. You, M. Zhang, L. Zhang, X. Zhou, K. Shi, and S. Gu, "Consistency trajectory matching for one-step generative super-resolution," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2025, pp. 12747–12756.
[76] K. Zhang et al., "NTIRE 2020 challenge on perceptual extreme super-resolution: Methods and results," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2020, pp. 2045–2057.
[77] C. Lyu et al., "RTMDet: An empirical study of designing real-time object detectors," 2022, *arXiv:2212.07784*.
[78] K. Chen, B. Chen, C. Liu, W. Li, Z. Zou, and Z. Shi, "RSMamba: Remote sensing image classification with state space model," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.
[79] W. Peebles and S. Xie, "Scalable diffusion models with transformers," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 4195–4205.



**Ce Wang** received the B.S. and M.S. degrees from the School of Remote Sensing and Information Engineering, Wuhan University, Wuhan, China, in 2024 and 2026, respectively.

His research interests include remote sensing image processing, image super-resolution, and diffusion-based generative models.



**Zhenyu Hu** received the M.S. degree from the School of Remote Sensing and Information Engineering, Wuhan University, Wuhan, China, in 2025.

His research interests include image super-resolution, computer vision, and deep learning.



**Wanjie Sun** received the Ph.D. degree from Wuhan University, Wuhan, China, in 2020.

He is currently an Associate Research Fellow with the School of Remote Sensing and Information Engineering, Wuhan University. His research interests include image processing, computer vision, and remote sensing, with emphasis on super-resolution and image enhancement.

Dr. Sun was a recipient of the National Postdoctoral Program for Innovative Talents.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 14:26:39 UTC from IEEE Xplore. Restrictions apply.